MIL-M-28787/304A(EC)
10 July 1981
SUPERSEDING
MIL-M-28787/304(EC)
25 September 1980

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MICROPROGRAM CONTROLLER, KEY CODE ZJU

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic microprogram controller module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicition, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, Naval Electronic Systems Command, ATTN ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.686 failures per million operating hours for class I or 2.242 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be ZJU.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/304-1 for class I or M28787/304-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2)

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 $\pm$0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc $\pm$1 percent

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures The module shall be tested as specified in table I.

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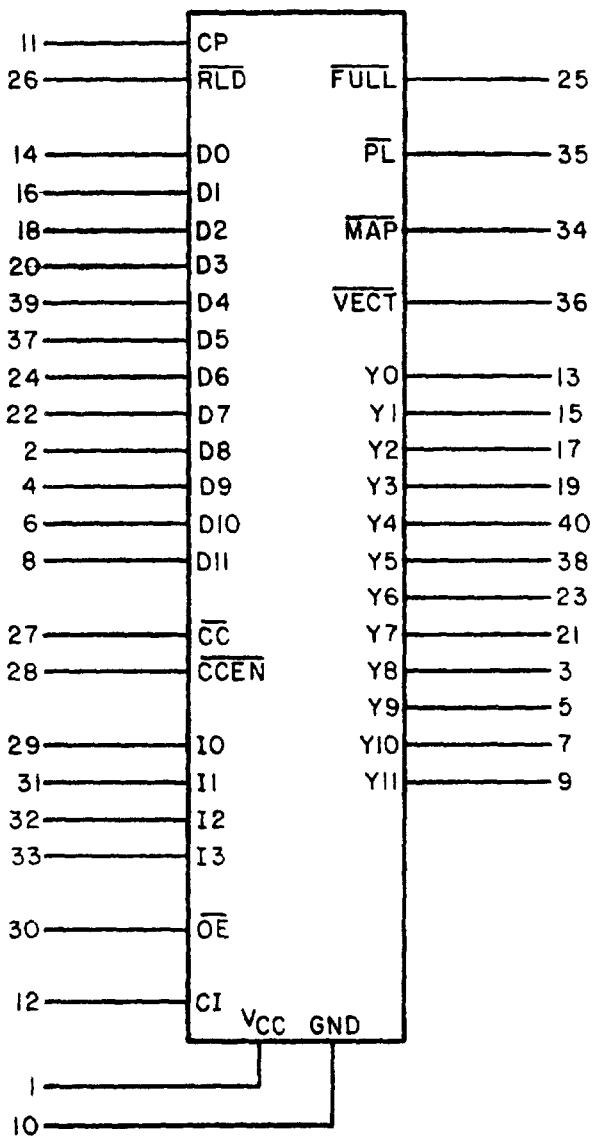


FIGURE 1. ZJU module function diagram, microprogram controller

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III		1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B	Voltage stress check	IV	S	← Module must pass remaining tests →										
C1-C22	Input breakdown current	V	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
D1-D22	Input clamp diode voltage	VI	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
E1-E16	Output short circuit current	VII	X	-30	-85	-30	-85	-30	-85	-30	-85	-30	-85	mA
F1-F12	Input current, high level	VIII	X	0	80	0	80	0	80	0	80	0	80	µA
F13,F14	Input current, high level	VIII	X	0	30	0	30	0	30	0	30	0	30	µA
F15-F20	Input current, high level	VIII	X	0	40	0	40	0	40	0	40	0	40	µA
F21	Input current, high level	VIII	X	0	50	0	50	0	50	0	50	0	50	µA
F22	Input current, high level	VIII	X	0	100	0	100	0	100	0	100	0	100	µA
G1-G12	Input current, low level	IX	X	0.0	-0.96	0.0	-0.96	0.0	-0.96	0.0	-0.96	0.0	-0.96	mA
G13 G14	Input current, low level	IX	X	0.0	-0.60	0.0	-0.60	0.0	-0.60	0.0	-0.60	0.0	-0.60	mA
G15-G20	Input current, low level	IX	X	0.0	-0.80	0.0	-0.80	0.0	-0.80	0.0	-0.80	0.0	-0.80	mA
G21	Input current, low level	IX	X	0.0	-1.45	0.0	-1.45	0.0	-1.45	0.0	-1.45	0.0	-1.45	mA
G22	Input current, low level	IX	X	0.0	-2.36	0.0	-2.36	0.0	-2.36	0.0	-2.36	0.0	-2.36	mA
H1-H16	Output voltage, high level	X	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J16	Output voltage, low level	XI	X	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	V
K1-K12	High impedance leakage current, high level	XII	X	-1	50	-1	50	-1	50	-1	50	-1	50	µA
L1-L12	High impedance leakage current, low level	XIII	X	1	-50	1	-50	1	-50	1	-50	1	-50	µA
M	Power supply current	XIV	X	28	287	30	303	30	303	34	340	34	340	mA

See footnotes at end of table

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TABLE I Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
N1-N10	Dynamic functional test	XV	X	◄ See table XV ►										
P1-P16	Transition time (t_{TLH}), C_L = 50 pF	XVI	S	1	17	1	17	1	19	1	17	1	21	ns
Q1-Q16	Transition time (t_{THL}), C_L = 50 pF	XVII	S	1	10	1	10	1	12	1	10	1	14	ns
R1-R12	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	2	18	2	18	2	21	2	18	2	25	ns
R13-R24	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	5	52	5	52	6	63	5	52	7	75	ns
R25-R27	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	3	37	3	37	4	47	3	37	5	58	ns
R28-R39	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	3	30	3	30	3	39	3	30	4	48	ns
R40-R51	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	3	32	3	32	4	41	3	32	5	50	ns
R52	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	4	42	4	42	5	54	4	42	6	67	ns
R53-R64	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	3	38	3	38	4	49	3	38	6	61	ns
S1-S12	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	S	2	18	2	18	2	21	2	18	2	25	ns
S13-S24	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	S	5	52	5	52	6	63	5	52	7	75	ns
S25-S27	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	S	3	37	3	37	4	47	3	37	5	58	ns
S28-S39	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	S	3	30	3	30	3	39	3	30	4	48	ns
S40-S51	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	S	3	32	3	32	4	41	3	32	5	50	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
S52	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	S'	4	42	4	42	5	54	4	42	6	67	ns
S53-S64	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	S	3	38	3	38	4	49	3	38	6	61	ns
T1-T12	Disable delay time(t_{PLZ}), C_L = 50 pF	XX	S	5	55	5	55	6	65	5	55	7	75	ns
U1-U12	Disable delay time (t_{PHZ}), C_L = 50 pF	XXI	S	6	60	6	60	7	70	6	60	8	80	ns
V1-V12	Enable recovery time (t_{PZL}), C_L = 50 pF	XXII	S	5	55	5	55	6	65	5	55	7	75	ns
W1-W12	Enable recovery time (t_{PZH}), C_L = 50 pF	XXIII	S	6	60	6	60	7	70	6	60	8	80	ns
X1-X16	Output crosstalk, low level	XXIV	Q	0	1,000	0	1,000	0	1,000	0	1,000	0	1,000	mV
Y1-Y16	Output crosstalk, high level	XXV	Q	0	500	0	500	0	500	0	500	0	500	mV
Z	Power supply transient amplitude	XXVI	Q	0	400	0	400	0	400	0	400	0	400	mV
AA	Power supply filter capacitance	XXVII	S	3.15	5.85	3.15	5.85	3.15	5.85	3.15	5.85	3.15	5.85	uF
BB1-BB22	Input capacitance	XXVIII	Q	2	30	2	30	2	30	2	30	2	30	pF
CC1-CC16	Output capacitance	XXIX	Q	2	30	2	30	2	30	2	30	2	30	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through CC shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows. A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through CC in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through AA shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition AA, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through AA shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through CC shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787

6 NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Module description. The following electrical parameters reflect the maximum case temperature end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation (outputs not loaded)	1.87	1.87	W
Input voltage:			
Low level	-0.5 min, 0.8 max	-0.5 min, 0.8 max	V
High level	2.0 min, V_{CC} max	2.0 min, V_{CC} max	V

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		Class I	Class II	Unit
Setup time, relative to rising edge of clock				
D0-D11	R	28 min	28 min	ns
D0-D11	PC	62 min	62 min	ns
I0-I3		110 min	110 min	ns
$\overline{CC}$		86 min	86 min	ns
$\overline{CCEN}$		86 min	86 min	ns
$\overline{CI}$		58 min	58 min	ns
$\overline{RLD}$		42 min	42 min	ns
Holdtime, relative to rising edge of clock·				
D0-D11	R	6 min	6 min	ns
D0-D11	PC	4 min	4 min	ns
I0-I3		0 min	0 min	ns
$\overline{CC}$		0 min	0 min	ns
$\overline{CCEN}$		0 min	0 min	ns
$\overline{CI}$		5 min	5 min	ns
$\overline{RLD}$		6 min	6 min	ns
Minimum clock low time		58 min	58 min	ns
Minimum clock high time		42 min	42 min	ns
Minimum clock period (Inputs = 8, 9, 15)		143 min	143 min	ns
Minimum clock period (all other inputs)		100 min	100 min	ns

6.5 Module function. This module is an address sequencer intended for controlling the sequences of execution of microinstructions stored in the microprogram memory. Besides the capability of sequential access, it provides conditional branching to any microinstruction within its 4096 microword range. A last-in, first-out stack provides microsubroutine return linkage and looping capability, there are 5 levels of nesting of microsubroutines. Microinstruction loop count control is provided with a count capacity of 4096. During each microinstruction, the module provides a 12-bit address from one of four sources

a. The microprogram address register (=PC), which usually contains an address one greater than the previous address
b. An external (direct) input (D)
c. A register/counter (R) retaining data loaded during a previous microinstruction
d. A five-deep last-in, first-out stack (F)

6.6 Module use. The ERD, ZJU, ZJX, and MJC modules are a series of designs implementing the 2900 family as building blocks to emulate microprogrammed computers or controllers.

ERD = Microprocessor.
ZJU = Microprogram controller.
ZJX = Interrupt controller.
MJC = Sequencer.

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6.7 Pin functions. Pin functions for the module are as follows:

Abbreviation	Name	Pin numbers	Function
D0-D11	Direct Input Bit 0-11	14,16,18, 20,39,37, 24,22,2,4, 6,8	Direct input to register/counter and multiplexer.
I0-I3	Instruction Bit 0-3	29,31,32, 33	Selects one-of-sixteen instructions
$\overline{CC}$	Condition Code	27	Used as test criterion. Pass test is a LOW on $\overline{CC}$.
$\overline{CCEN}$	Condition Code Enable	28	Whenever the signal is HIGH, $\overline{CC}$ is ignored and the part operates as though $\overline{CC}$ were true (LOW).
CI	Carry-In	12	Low order carry input to incrementer for microprogram counter.
$\overline{RLD}$	Register Load	26	When LOW forces loading of register/counter regardless of instruction or condition.
$\overline{OE}$	Output Enable	30	Three-state control of Y outputs.
CP	Clock Pulse	11	Triggers all internal state changes at LOW-to-HIGH edge.
Y0-Y11	Microprogram Address Bit 0-11	13,15,17, 19,40,38, 23,21,3,5, 7,9	Address to microprogram memory
$\overline{FULL}$	Full	25	Indicates that five items are on the stack
$\overline{PL}$	Pipeline Address	35	Can select No. 1 source (usually Pipeline Register) as direct input sources.
$\overline{MAP}$	Map Address	34	Can select No. 2 source (usually Mapping PROM or PLA) as direct input source.
$\overline{VECT}$	Vector Address	36	Can select No. 3 source (for example, Interrupt Starting Address) as direct input source.

6.8 Instructions. See table II for operation of the instruction bit I0-I3.

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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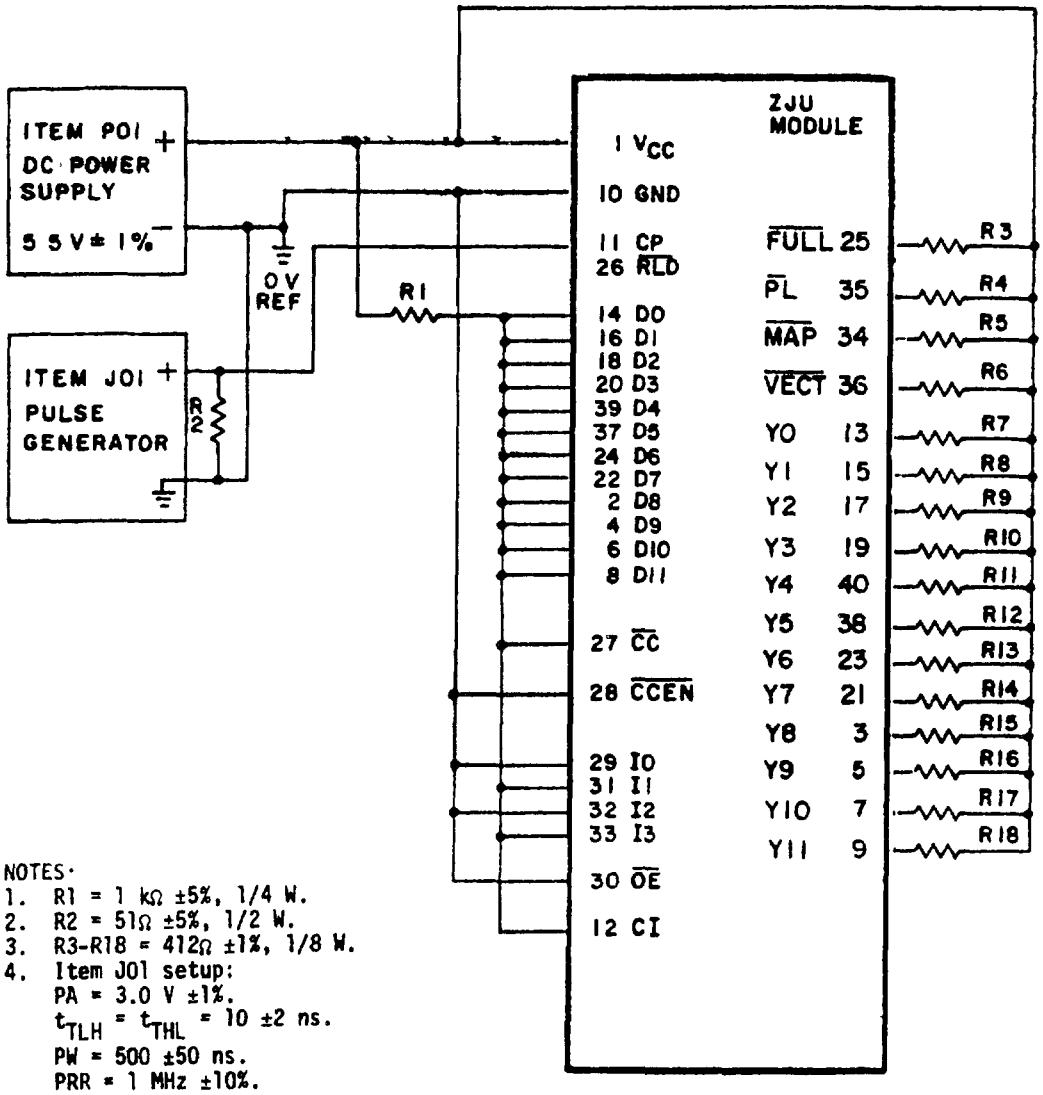
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TABLE II. Instructions.

I3-I0	Mnemonic	Name	Register/ Counter contents	$\overline{CCEN}$ = LOW and $\overline{CC}$ = HIGH		$\overline{CCEN}$ = HIGH or $\overline{CC}$ = LOW		Register/ Counter	Enable
				Y	Stack	Y	Stack		
0	JZ	JUMP ZERO	X	0	CLEAR	0	CLEAR	HOLD	PL
1	CJS	COND JSB PL	X	PC	HOLD	D	PUSH	HOLD	PL
2	JMAP	JUMP MAP	X	D	HOLD	D	HOLD	HOLD	$\overline{MAP}$
3	CJP	COND JUMP PL	X	PC	HOLD	D	HOLD	HOLD	PL
4	PUSH	PUSH/COND LD CNTR	X	PC	PUSH	PC	PUSH	1/	PL
5	JSRP	COND JSB R/PL	X	R	PUSH	D	PUSH	HOLD	PL
6	CJV	COND JUMP VECTOR	X	PC	HOLD	D	HOLD	HOLD	$\overline{VECT}$
7	JRP	COND JUMP R/PL	X	R	HCLD	D	HOLD	HOLD	PL
8	RFCT	REPEAT LOOP, CNTR ≠0	≠0	F	HOLD	F	HOLD	DEC	PL
			=0	PC	POP	PC	POP	HOLD	PL
9	RPCT	REPEAT PL, CNTR ≠0	≠0	D	HOLD	D	HOLD	DEC	PL
			=0	PC	HOLD	PC	HOLD	HOLD	PL
10	CRTN	COND RTN	X	PC	HOLD	F	POP	HOLD	PL
11	CJPP	COND JUMP PL and POP	X	PC	HOLD	D	POP	HOLD	PL
12	LDCT	LD CNTR and CONTINUE	X	PC	HOLD	PC	HOLD	LOAD	PL
13	LOOP	TEST END LOOP	X	F	HOLD	PC	POP	HOLD	PL
14	CONT	CONTINUE	X	PC	HOLD	PC	HOLD	HOLD	PL
15	TWB	THREE-WAY BRANCH	≠0	F	HOLD	PC	POP	DEC	PL
			=0	D	POP	PC	POP	HOLD	PL

1/ If $\overline{CCEN}$ = LOW and $\overline{CC}$ = HIGH, hold, else load. X = Don't Care.

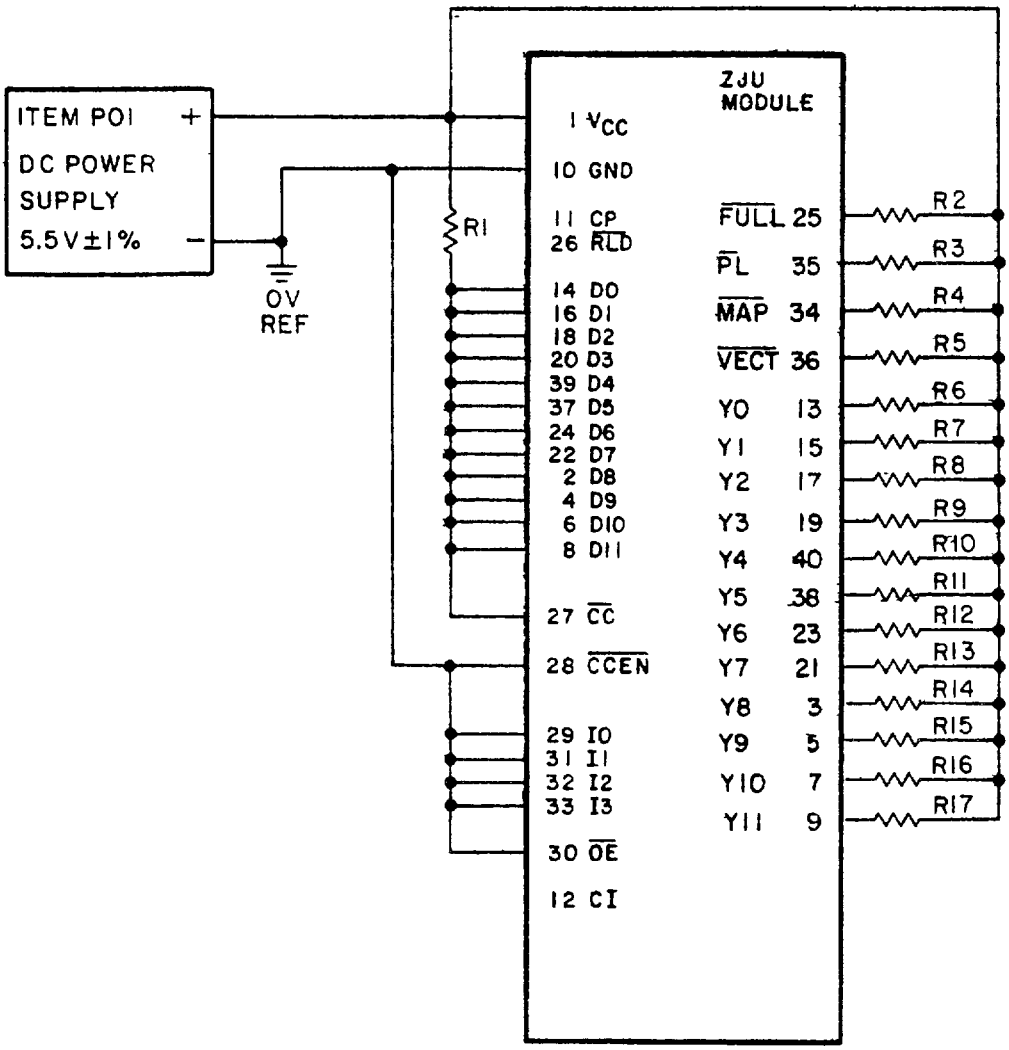
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NOTES:
1. R1 = 1 kΩ ±5%, 1/4 W.
2. R2 = 51Ω ±5%, 1/2 W.
3. R3-R18 = 412Ω ±1%, 1/8 W.
4. Item J01 setup:
   PA = 3.0 V ±1%.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   PW = 500 ±50 ns.
   PRR = 1 MHz ±10%.

FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1 = 1.0 kΩ ±5%, 1/4 W.
2 R2-R17 = 412Ω ±1%, 1/8 W

FIGURE 3 Component temperature test circuit

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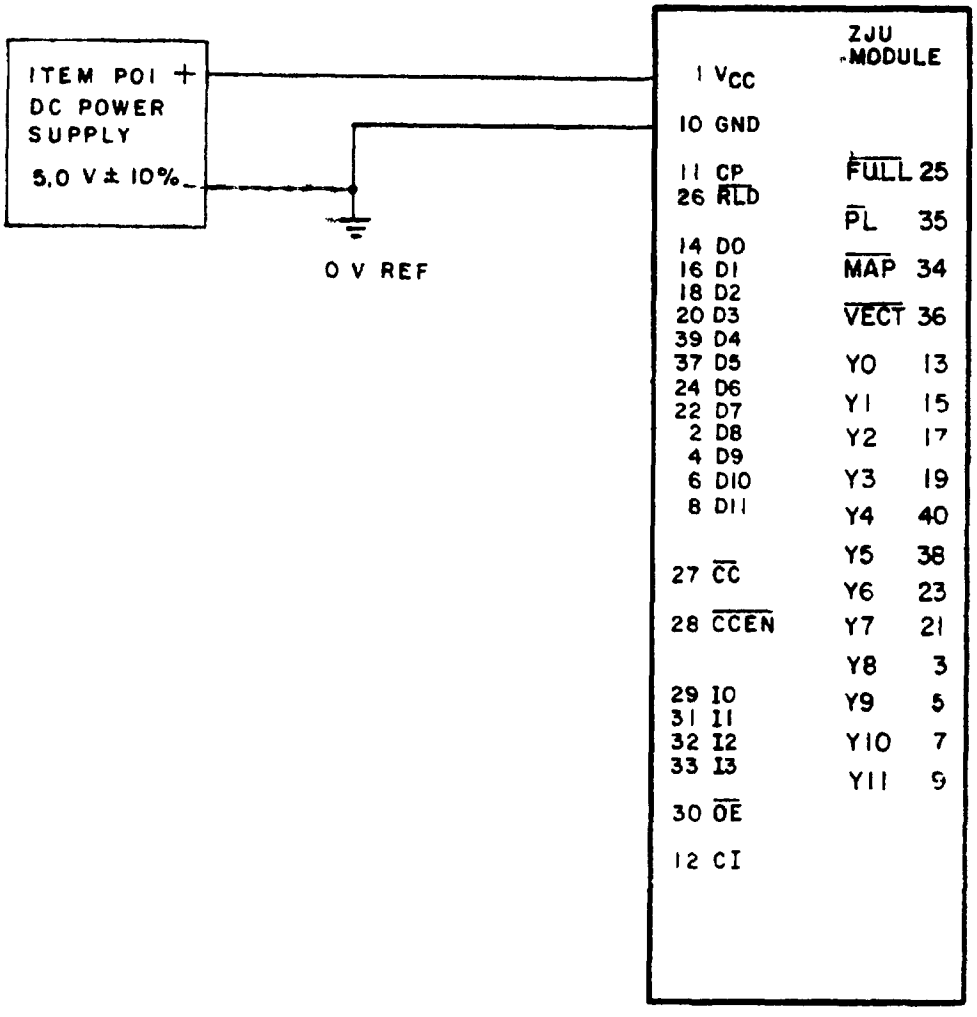


FIGURE 4. Vibration test circuit.

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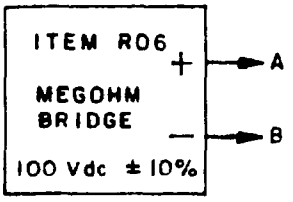
TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	All pins	Frame	10	∞	MΩ	5

TABLE IV. Voltage stress check test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
				Min	Max		
B	V_{CC} Voltage stress check	← Must pass remainder of tests →					6

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WARNING:

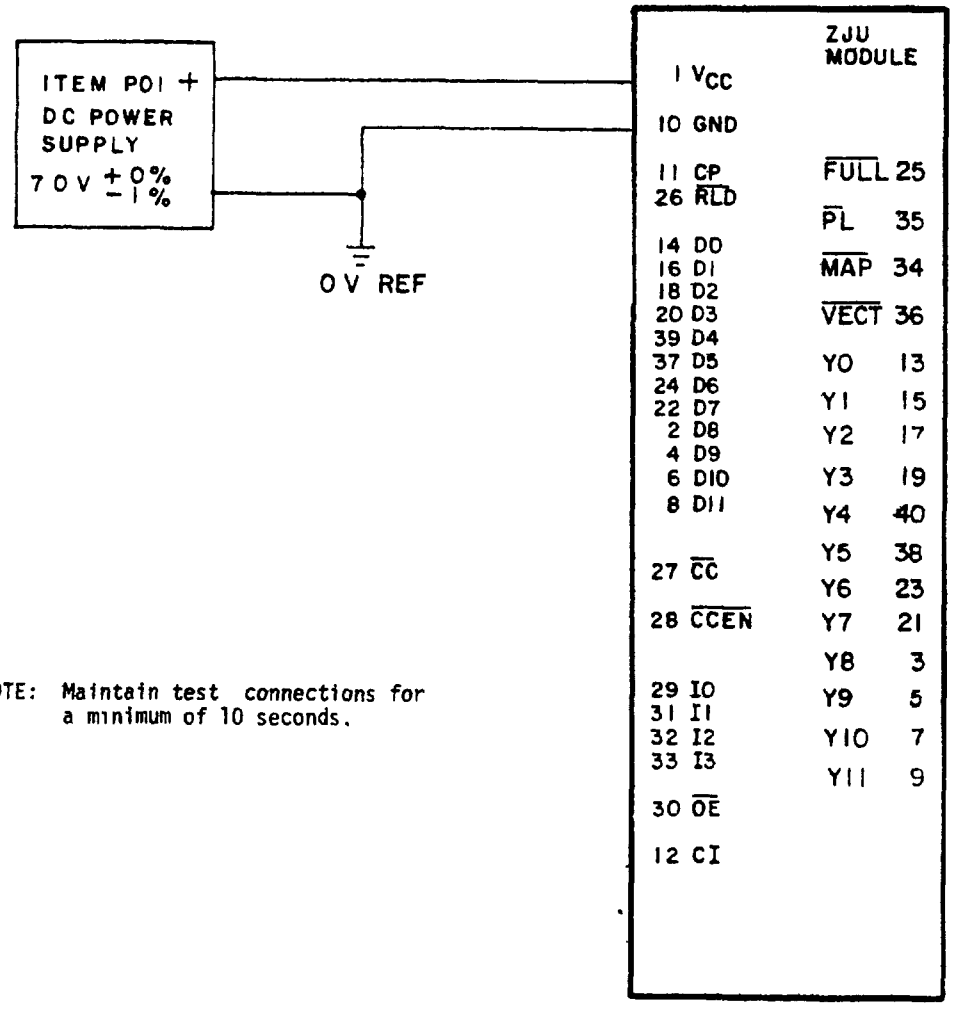
When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

	ZJU MODULE	
1	V_{CC}	
10	GND	
11	CP	$\overline{FULL}$ 25
26	$\overline{RLD}$	
		$\overline{PL}$ 35
14	D0	
16	D1	$\overline{MAP}$ 34
18	D2	
20	D3	$\overline{VECT}$ 36
39	D4	
37	D5	Y0 13
24	D6	
22	D7	Y1 15
2	D8	Y2 17
4	D9	
6	D10	Y3 19
8	D11	Y4 40
		Y5 38
27	$\overline{CC}$	Y6 23
28	$\overline{CCEN}$	Y7 21
		Y8 3
29	I0	Y9 5
31	I1	
32	I2	Y10 7
33	I3	Y11 9
30	$\overline{OE}$	
12	CI	

FIGURE 5. Isolation test circuit for table III.

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NOTE: Maintain test connections for a minimum of 10 seconds.

FIGURE 6. Voltage stress check test circuit for table IV

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TABLE V. Input breakdown current test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
C1	CP	11	2,4,6,8,12,14,16,18,20,22,24,26,27,28,29,30,31,32,33,37,39	0.0	1.0	mA	7
C2	$\overline{RLD}$	26	2,4,6,8,11,12,14,16,18,20,22,24,27,28,29,30,31,32,33,37,39				
C3	D0	14	2,4,6,8,11,12,16,18,20,22,24,26,27,28,29,30,31,32,33,37,39				
C4	D1	16	2,4,6,8,11,12,14,18,20,22,24,26,27,28,29,30,31,32,33,37,39				
C5	D2	18	2,4,6,8,11,12,14,16,20,22,24,26,27,28,29,30,31,32,33,37,39				
C6	D3	20	2,4,6,8,11,12,14,16,18,22,24,26,27,28,29,30,31,32,33,37,39				
C7	D4	39	2,4,6,8,11,12,14,16,18,20,22,24,26,27,28,29,30,31,32,33,37				
C8	D5	37	2,4,6,8,11,12,14,16,18,20,22,24,26,27,28,29,30,31,32,33,39				
C9	D6	24	2,4,6,8,11,12,14,16,18,20,22,26,27,28,29,30,31,32,33,37,39				

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TABLE V. Input breakdown current test - Continued.

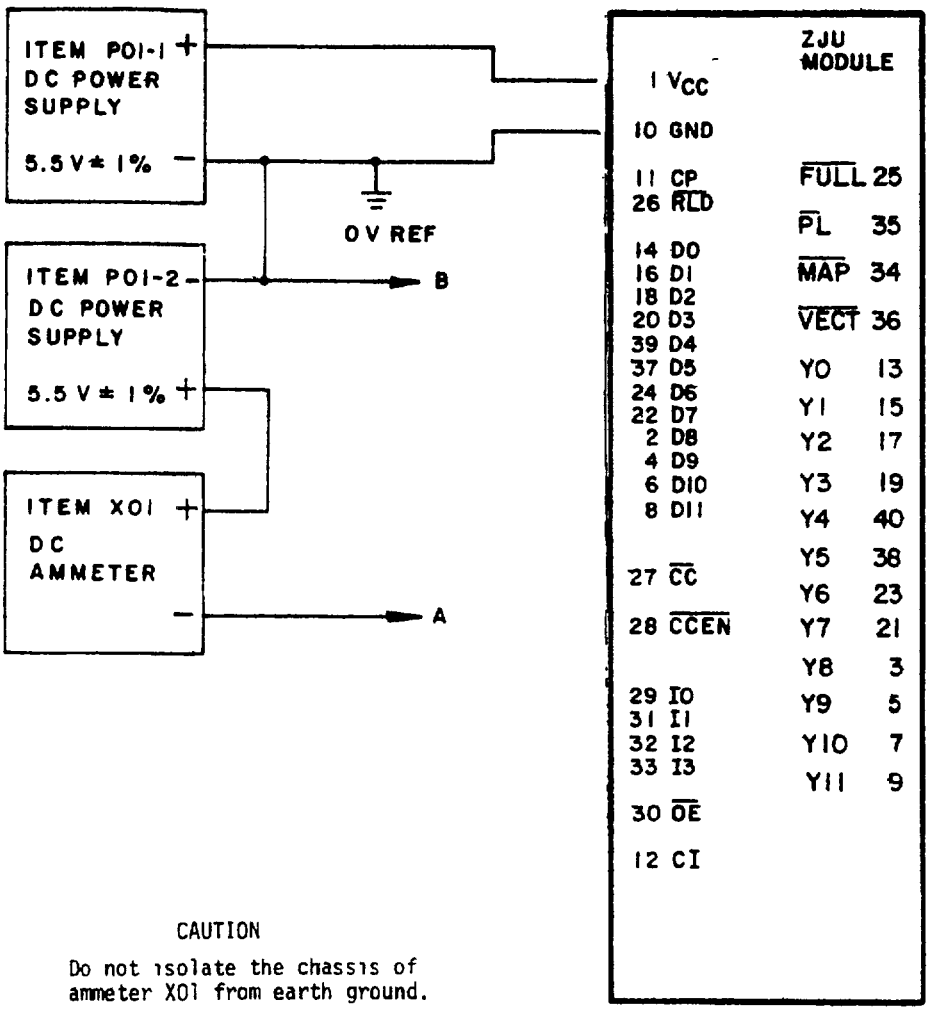
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C10	D7 Input breakdown current	22	2,4,6,8,11,12,14,16,18, 20,24,26,27,28,29,30, 31,32,33,37,39	0.0	1.0	mA	7
C11	D8	2	4,6,8,11,12,14,16,18, 20,22,24,26,27,28,29, 30,31,32,33,37,39				
C12	D9	4	2,6,8,11,12,14,16,18, 20,22,24,26,27,28,29, 30,31,32,33,37,39				
C13	D10	6	2,4,8,11,12,14,16,18, 20,22,24,26,27,28,29, 30,31,32,33,37,39				
C14	D11	8	2,4,6,11,12,14,16,18, 20,22,24,26,27,28,29, 30,31,32,33,37,39				
C15	$\overline{CC}$	27	2,4,6,8,11,12,14,16,18, 20,22,24,26,28,29,30, 31,32,33,37,39				
C16	$\overline{CCEN}$	28	2,4,6,8,11,12,14,16,18, 20,22,24,26,27,29,30, 31,32,33,37,39				
C17	I0	29	2,4,6,8,11,12,14,16,18, 20,22,24,26,27,28,30, 31,32,33,37,39				
C18	I1	31	2,4,6,8,11,12,14,16,18, 20,22,24,26,27,28,29, 30,32,33,37,39				

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter		Connect		Initial limits 25°+5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C19	I2	Input breakdown current	32	2,4,6,8,11,12,14,16,18,20,22,24,26,27,28,29,30,31,33,37,39	0.0	1.0	mA	7
C20	I3		33	2,4,6,8,11,12,14,16,18,20,22,24,26,27,28,29,30,31,32,37,39				
C21	$\overline{OE}$		30	2,4,6,8,11,12,14,16,18,20,22,24,26,27,28,29,31,32,33,37,39				
C22	CI		12	2,4,6,8,11,14,16,18,20,22,24,26,27,28,29,30,31,32,33,37,39				

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CAUTION

Do not isolate the chassis of ammeter XO1 from earth ground.

FIGURE 7 Input breakdown current test circuit for table V

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TABLE VI. Input clamp diode voltage test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
D1	CP	11	-0.4	-1.5	V	8
D2	$\overline{RLD}$	26				
D3	D0	14				
D4	D1	16				
D5	D2	18				
D6	D3	20				
D7	D4	39				
D8	D5	37				
D9	D6	24				
D10	D7	22				
D11	D8	2				
D12	D9	4				
D13	D10	6				
D14	D11	8				
D15	$\overline{CC}$	27				
D16	$\overline{CCEN}$	28				
D17	I0	29				
D18	I1	31				

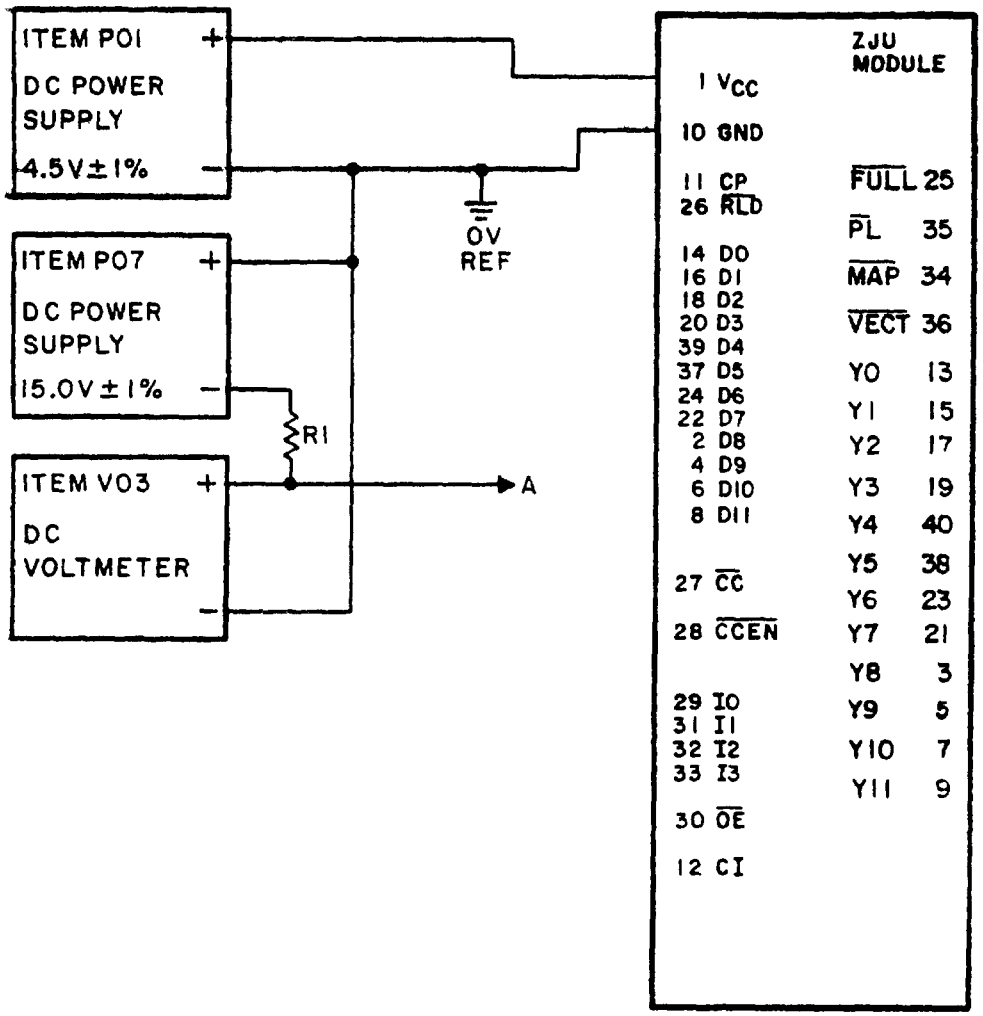
Parameter: Input clamp diode voltage

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TABLE VI. Input clamp diode voltage test - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
D19	I2	Input clamp diode voltage	32	-0.4	-1.5	V	8
D20	I3	↓	33	↓	↓	↓	↓
D21	$\overline{OE}$		30				
D22	CI		12				

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NOTE: R1 = 750Ω ±1%, 1/4 W.

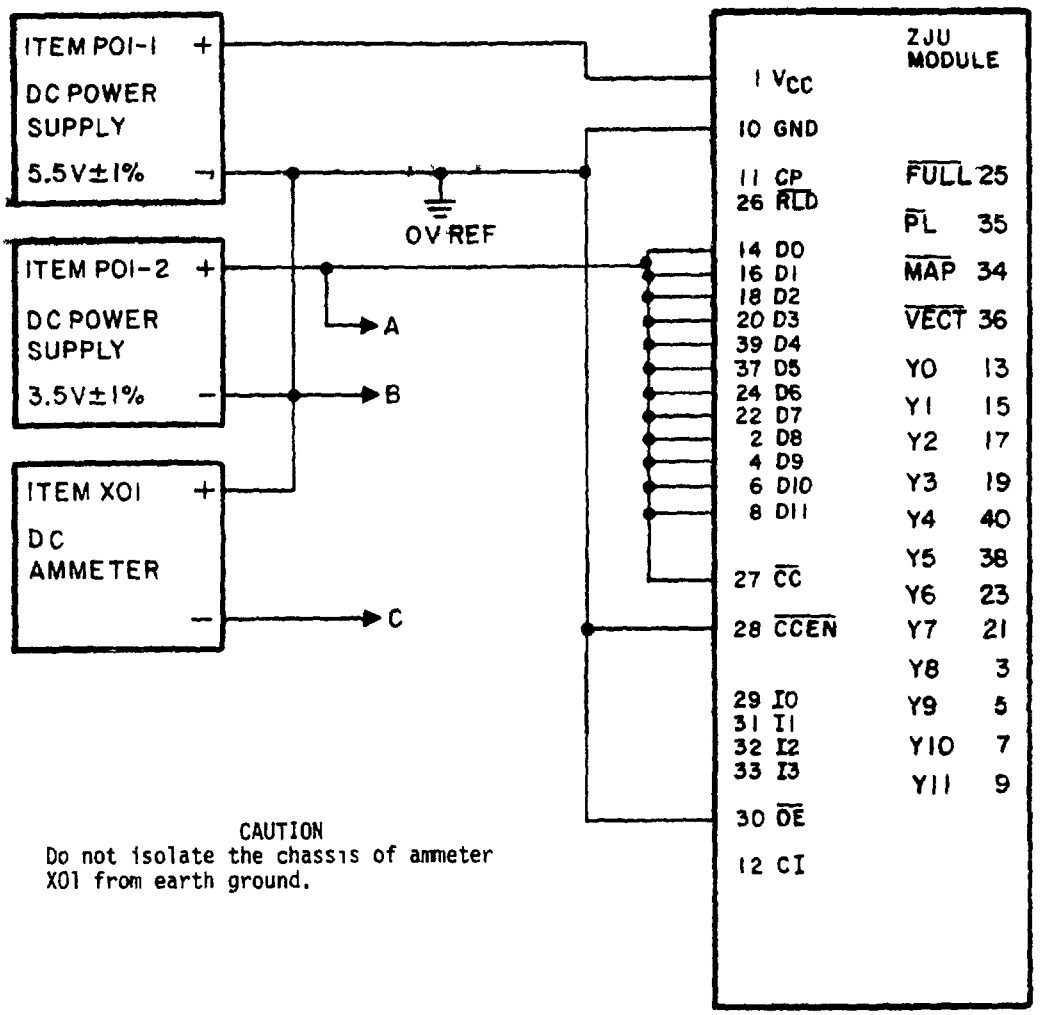
FIGURE 8. Input clamp diode voltage test circuit for table VI.

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TABLE VII. Output short circuit current test.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
E1	YO	Output short circuit current	31	29,32,33	13	-30	-85	mA	9
E2	Y1				15				
E3	Y2				17				
E4	Y3				19				
E5	Y4				40				
E6	Y5				38				
E7	Y6				23				
E8	Y7				21				
E9	Y8				3				
E10	Y9				5				
E11	Y10				7				
E12	Y11				9				
E13	$\overline{\text{FULL}}$				25				
E14	$\overline{\text{PL}}$				35				
E15	$\overline{\text{VECT}}$				36				
E16	$\overline{\text{MAP}}$		Open	29,31,32, 33	34				

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CAUTION
Do not isolate the chassis of ammeter XOI from earth ground.

NOTE: Duration of test shall not exceed 1 second.

FIGURE 9. Output short circuit current test circuit for table VII.

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TABLE VIII. Input current (high-level) test.

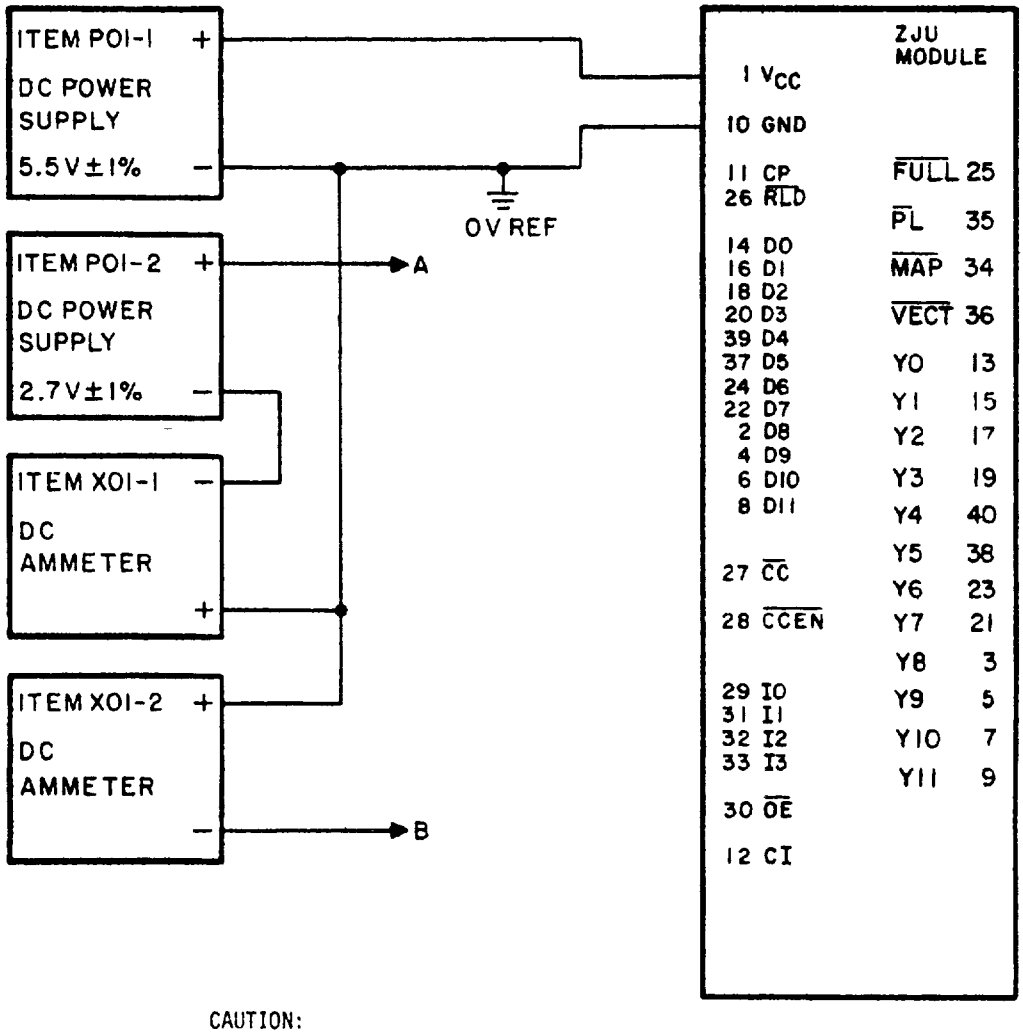
Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
F1	D0	Input current, high level	14	Open	0	80	µA	10
F2	D1		16					
F3	D2		18					
F4	D3		20					
F5	D4		39					
F6	D5		37					
F7	D6		24					
F8	D7		22					
F9	D8		2					
F10	D9		4					
F11	D10		6					
F12	D11		8					
F13	CI		12			30		
F14	$\overline{\text{CCEN}}$		28			30		
F15	IO		29			40		
F16	I1		31					
F17	I2		32					
F18	I3		33					
F19	$\overline{\text{OE}}$		30					
F20	$\overline{\text{RLD}}$		29					
F21	$\overline{\text{CC}}$		27			50		
F22	CP		11			100		

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TABLE IX. Input current (low-level) test.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
G1	D0	Input current, low level	Open	14	0.0	-0.96	mA	10
G2	D1			16				
G3	D2			18				
G4	D3			20				
G5	D4			39				
G6	D5			37				
G7	D6			24				
G8	D7			22				
G9	D8			2				
G10	D9			4				
G11	D10			6				
G12	D11			8				
G13	CI			12		-0.60		
G14	$\overline{\text{CCEN}}$			28		-0.60		
G15	I0			29		-0.80		
G16	I1			31				
G17	I2			32				
G18	I3			33				
G19	$\overline{\text{OE}}$			30				
G20	$\overline{\text{RLD}}$			26				
G21	$\overline{\text{CC}}$			27		-1.45		
G22	CP			11		-2.36		

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTE Since items XO1-1 and XO1-2 are not used simultaneously, only one item XO1 is required.

FIGURE 10. Input current (high-level and low-level) test circuit for tables VIII and IX.

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TABLE X. Output voltage (high-level) test.

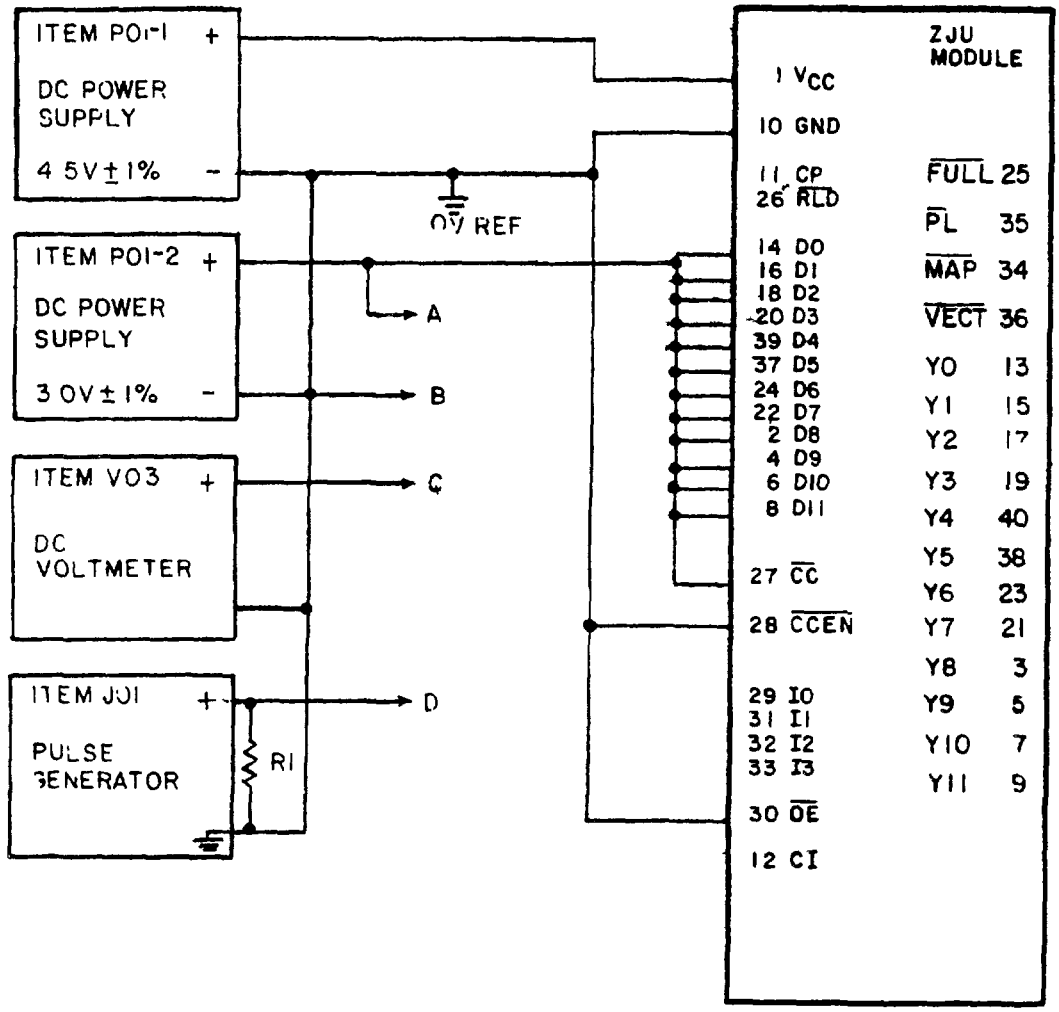
Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
H1	Y0	Output voltage, high level	31	29,32,33	13	Open	2.4	4.5	V	11
H2	Y1				15					
H3	Y2				17					
H4	Y3				19					
H5	Y4				40					
H6	Y5				38					
H7	Y6				23					
H8	Y7				21					
H9	Y8				3					
H10	Y9				5					
H11	Y10				7					
H12	Y11				9					
H13	$\overline{PL}$				35					
H14	$\overline{VECT}$				36					
H15	$\overline{MAP}$		Open	29,31,32,33	34					
H16	$\overline{FULL}$		Open	29,31,32,33	25					

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TABLE XI. Output voltage (low-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
J1	Y0 Output voltage, low level	Open	29,31,32,33	13	Open	0.0	0.5	V	11
J2	Y1			15					
J3	Y2			17					
J4	Y3			19					
J5	Y4			40					
J6	Y5			38					
J7	Y6			23					
J8	Y7			21					
J9	Y8			3					
J10	Y9			5					
J11	Y10			7					
J12	Y11			9					
J13	$\overline{PL}$			35					
J14	$\overline{VECT}$	31,32	29,33	36					
J15	$\overline{MAP}$	31	29,32,33	34					
J16	$\overline{FULL}$	32	29,31,33	25	11				11,12

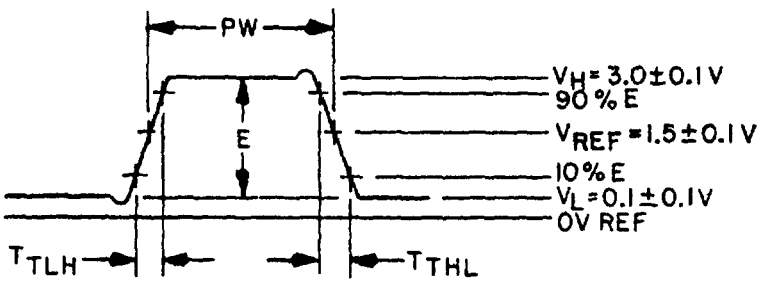
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:34Z
Check the source to verify that this is the current version before use.



NOTES
1. R1 = 51Ω ±5%, 1/2 W.
2. Load each output per figure 13.
3 Item J01 is used in single pulse mode. In test condition J16, apply 5 pulses as specified on figure 12.

FIGURE 11. Output voltage (high-level and low-level) test circuit for tables X and XI.

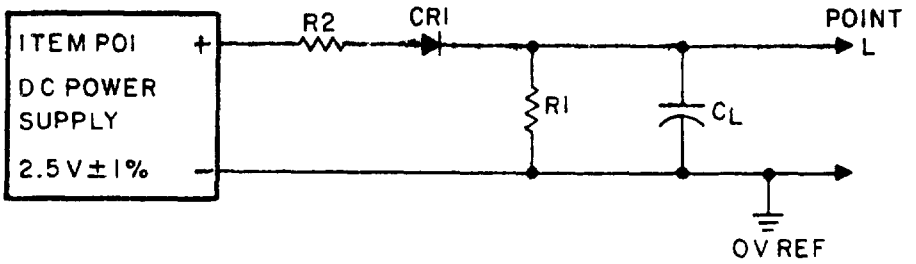
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:34Z
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NOTE Input pulse parameters:
$t_{TLH} = t_{THL}$ = 10 ±2 ns.
PW = 500 ±50 ns.

FIGURE 12. Output voltage (low-level) test circuit waveform for table XI.

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NOTES
1. CR1 = 1N4150 or equivalent.
2. C_L = 50 pF ±5% (including probe and parasitic capacitance).
3. R1 = 1.50 kΩ ±1%, 1/8 W.
4 R2 = Select value for a current flow of 12 mA ±5% out of point L with point L held at 0 5 V for output pins 13, 15, 17, 19, 40, 38, 23, 21, 3, 5, 7 and 9
5 R2 = Select value for a current flow of 8 mA ±5% out of point L with point L held at 0.5 V for output pins 35, 36, 34 and 25.
6. Capacitor C_L is not required for dc test parameters

FIGURE 13. ZJU logic load circuit, microprogram controller.

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TABLE XII. High impedance leakage current (high-level) test.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
K1	Y0	High impedance leakage current, high level	30	13	Open	-1	50	µA	14
K2	Y1			15					
K3	Y2			17					
K4	Y3			19					
K5	Y4			40					
K6	Y5			38					
K7	Y6			23					
K8	Y7			21					
K9	Y8			3					
K10	Y9			5					
K11	Y10			7					
K12	Y11			9					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:34Z
Check the source to verify that this is the current version before use.

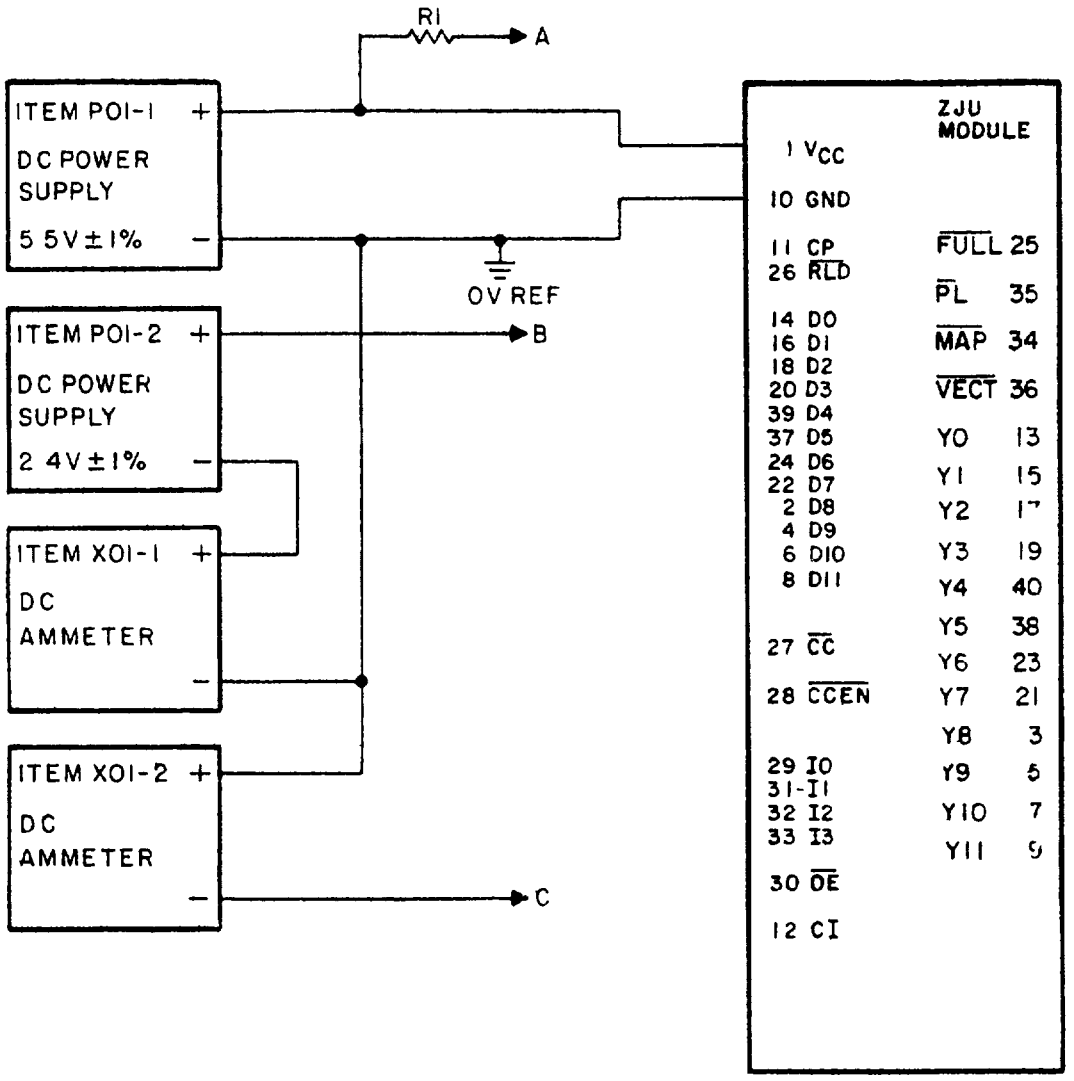
TABLE XIII. High impedance leakage current (low-level) test.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
L1	Y0	High impedance leakage current, low level	30	Open	13	1	-50	µA	14
L2	Y1				15				
L3	Y2				17				
L4	Y3				19				
L5	Y4				40				
L6	Y5				38				
L7	Y6				23				
L8	Y7				21				
L9	Y8				3				
L10	Y9				5				
L11	Y10				7				
L12	Y11				9				

TABLE XIV. Power supply current test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
M	Power supply current	27,28,29,30,31,32,33	28	287	mA	15

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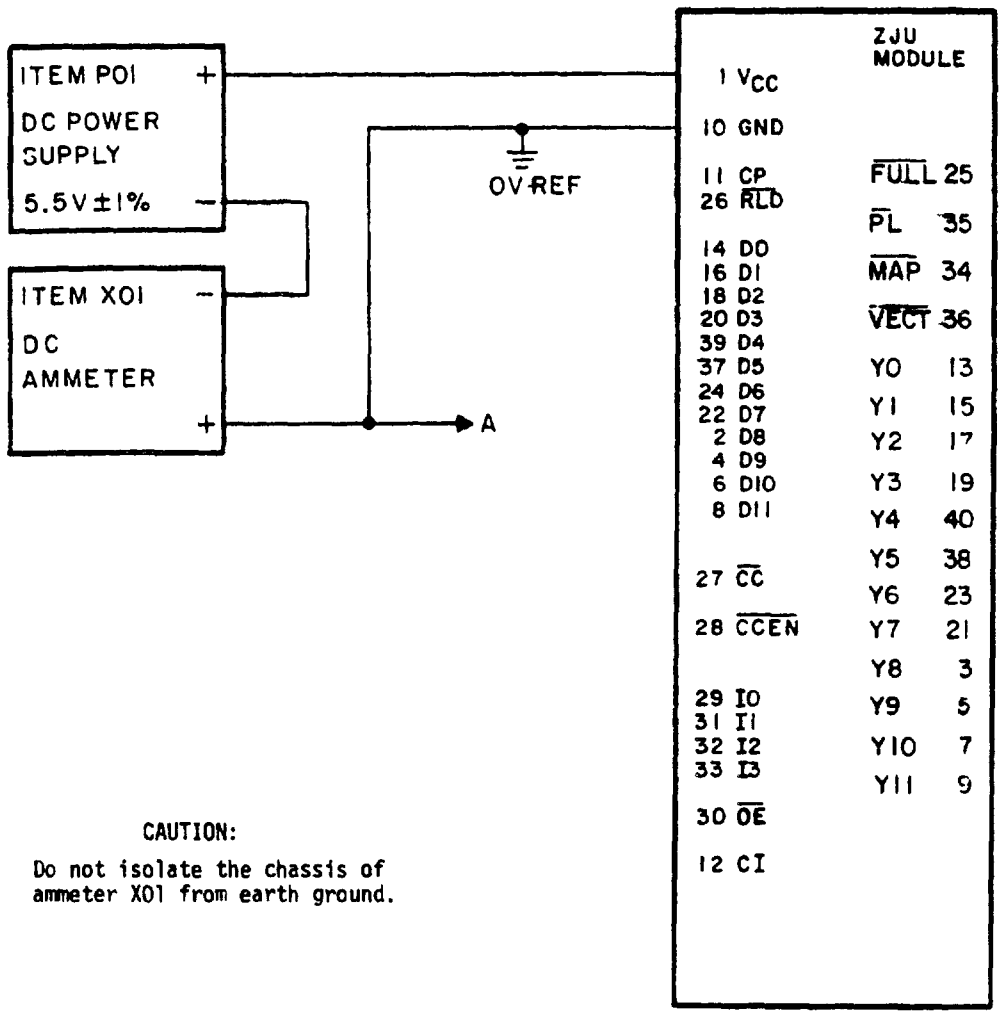
CAUTION

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES
1. R1 = 1.0 kΩ ±1%, 1/4 W.
2. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required

FIGURE 14. High impedance leakage current (high-level and low-level) test circuit for tables XII and XIII.

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 15. Power supply current test circuit for table XIV.

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TABLE XV. Dynamic functional test.

Test condition	Test description	Footnote	Test figure
N1	Minimum clock low time (V_{CC} = 4.5 V)	1/	16
N2	Minimum clock low time (V_{CC} = 5.5 V)	2/	16
N3	Minimum clock high time (V_{CC} = 4.5 V)	3/	17
N4	Minimum clock high time (V_{CC} = 5.5 V)	4/	17
N5	Minimum clock period for I = 14 (V_{CC} = 4.5 V)	5/	18
N6	Minimum clock period for I = 14 (V_{CC} = 5.5 V)	6/	18
N7	Minimum clock period for I = 8 (V_{CC} = 4.5 V)	7/	19
N8	Minimum clock period for I = 8 (V_{CC} = 5.5 V)	8/	19
N9	Setup and hold times (V_{CC} = 4.5 V)	9/	20, 21
N10	Setup and hold times (V_{CC} = 5.5 V)	10/	20, 21

1/ Test condition N1 checks minimum clock low time.
FPAT 0: Pass 1 checks high→low setup times on
$\overline{CC}$, $\overline{CCEN}$
Data→PC
CI

Pass 2 checks setup times on:
$\overline{CC}$, $\overline{CCEN}$
$\overline{Data}$→PC
$\overline{RLD}$

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Footnotes - Continued

1/ - Continued

FPAT 1: Pass 1 checks low→high setup times on:
$\overline{CC}$, $\overline{CCEN}$
Data→PC
CI

Pass 2 checks low→high setup times on;
$\overline{CC}$, $\overline{CCEN}$
Data→PC
$\overline{RLD}$ (high→low)

2/ Test condition N2 same as 1, except V_{CC} = 5.5 V.
3/ Test condition N3 checks minimum clock high time, V_{CC} = 4.5 V. No setup times checked, uses FPAT 6.
4/ Test condition N4 same as 3, except V_{CC} = 5.5 V.
5/ Test condition N5 checks minimum clock period for I = 14, V_{CC} = 4.5 V, uses FPAT 3.
6/ Test condition N6 same as 5, except V_{CC} = 5.5 V.
7/ Test condition N7 checks minimum clock period for I = 8, V_{CC} = 4.5 V.
8/ Test condition N8 same as 7, except V_{CC} = 5.5 V.
9/ Test condition N9 uses:
FPAT 4 (V_{CC} = 4.5 V): Pass 1 vectors 1 through 8 check high→low setup times and low→high hold times for:
Data→R
$\overline{RLD}$

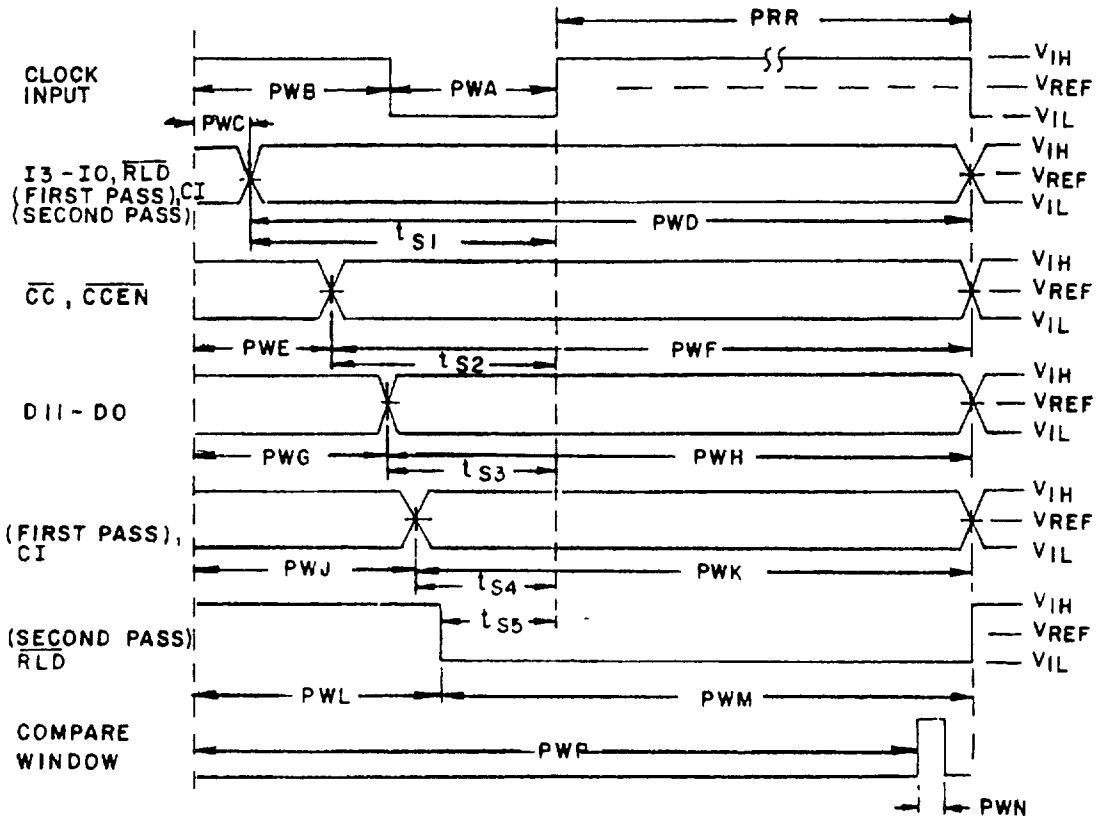
Pass 2 vectors 9 through 34 check low→high hold times for:
Data→PC
CI

FPAT 5. Pass 1 vectors 1 through 8 check:
Low→high setup times for Data→R
High→low setup time for $\overline{RLD}$
High→low hold time for D→R
Low→high hold time for $\overline{RLD}$

Pass 2 vectors 9 through 34 check High→low hold times for:
Data→PC
CI

10/ Test condition N10 same as 9, except V_{CC} = 5.5 V.

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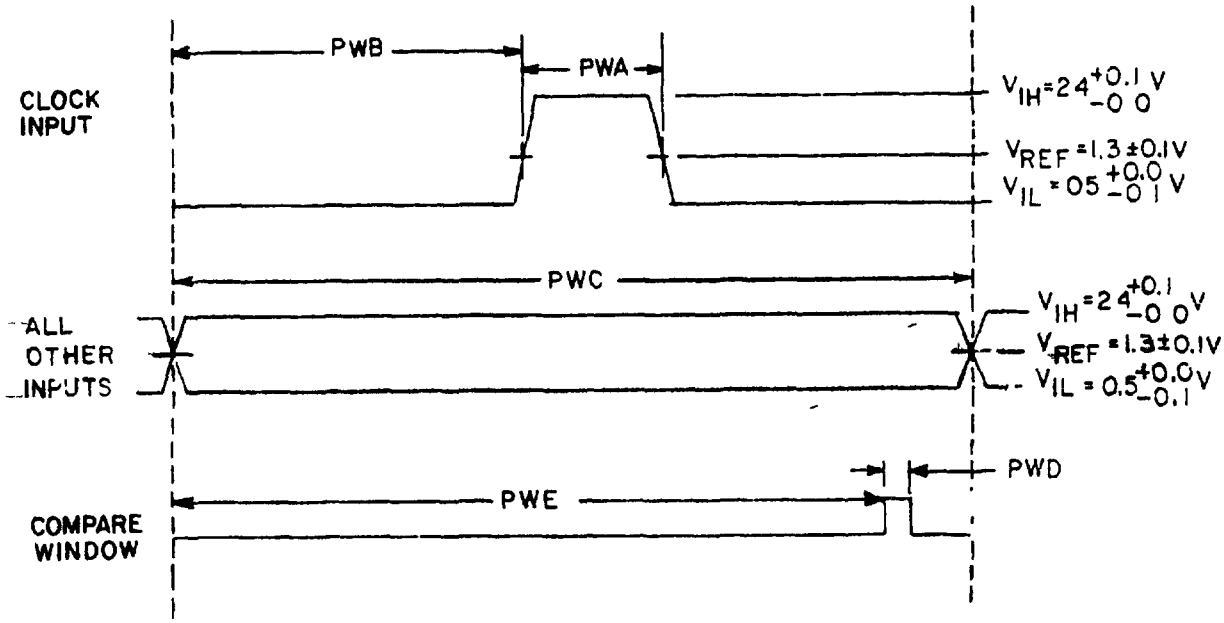


NOTE Input pulse parameters

V_{IH} = 2.4 +0 1, -0 0 V
V_{IL} = 0.5 +0 0, -0 1 V
V_{REF} = 1.3 ±0 1 V
PWA = 58 ±1 ns
PWB = 72 ±1 ns
PWC = 20 ±1 ns.
PWD = 284 ±1 ns
PWE = 44 ±1 ns
PWF = 260 ±1 ns.
PWG = 68 ±1 ns.
PWH = 236 ±1 ns.
PWJ = 72 ±1 ns.
PWK = 232 ±1 ns.
PWL = 88 ±1 ns
PWM = 216 ±1 ns
PWN = 10 ±1 ns
PWP = 280 ±1 ns
PRR = 304 ±1 ns
t_{S1} = 110 ±2 ns
t_{S2} = 86 ±2 ns
t_{S3} = 62 ±2 ns.
t_{S4} = 58 ±2 ns.
t_{S5} = 42 ±2 ns.

FIGURE 16 Dynamic functional test waveforms (N1, N2, FPAT0 and FPAT1) for table XV.

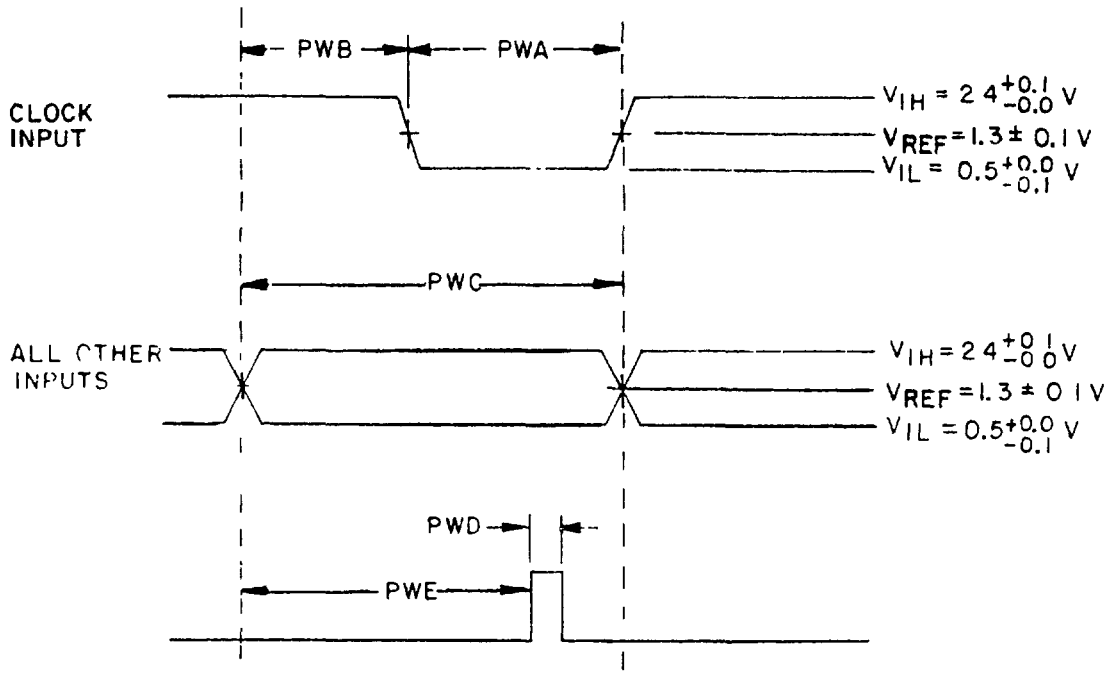
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NOTE: Input pulse parameters:
PWA = 42 ±1 ns.
PWB = 140 ±1 ns.
PWC = 304 ±1 ns.
PWD = 10 ±1 ns.
PWE = 280 ±1 ns.

FIGURE 17. Dynamic functional test waveforms (N3, N4 and FPAT6) for table XV.

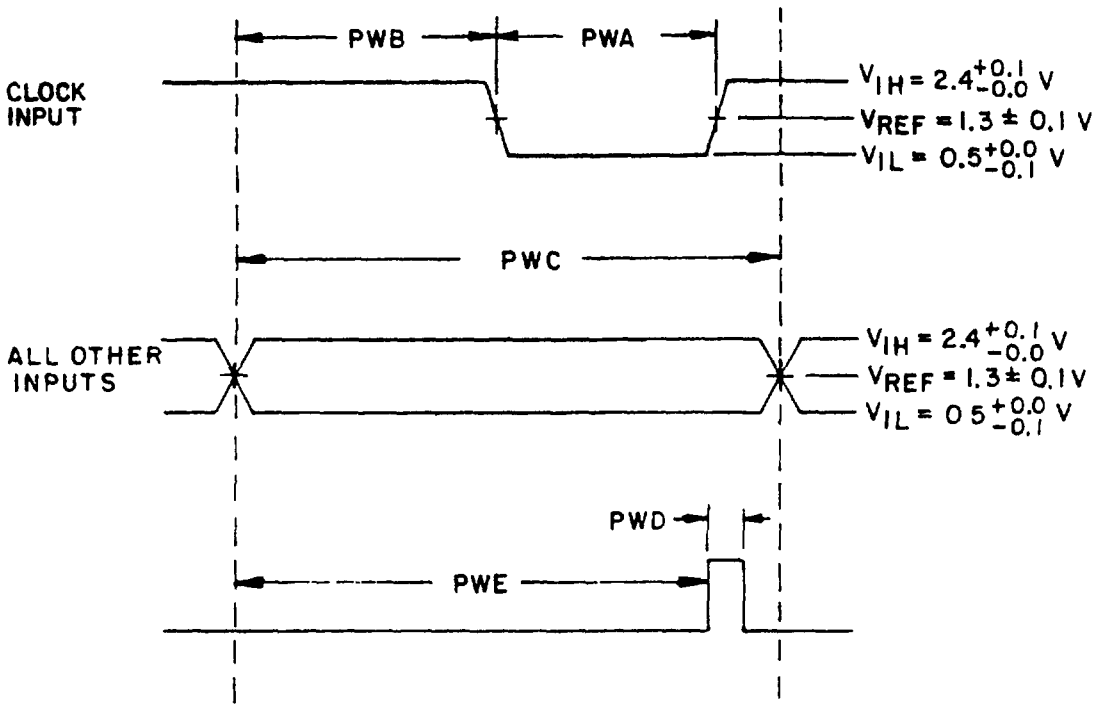
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:34Z
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NOTE Input pulse parameters
PWA = 60 ±1 ns
PWB = 44 ±1 ns
PWC = 104 ±1 ns.
PWD = 8 ±1 ns
PWE = 80 ±1 ns

FIGURE 18 Dynamic functional test waveforms (N5, N6 and FPAT3) for table XV.

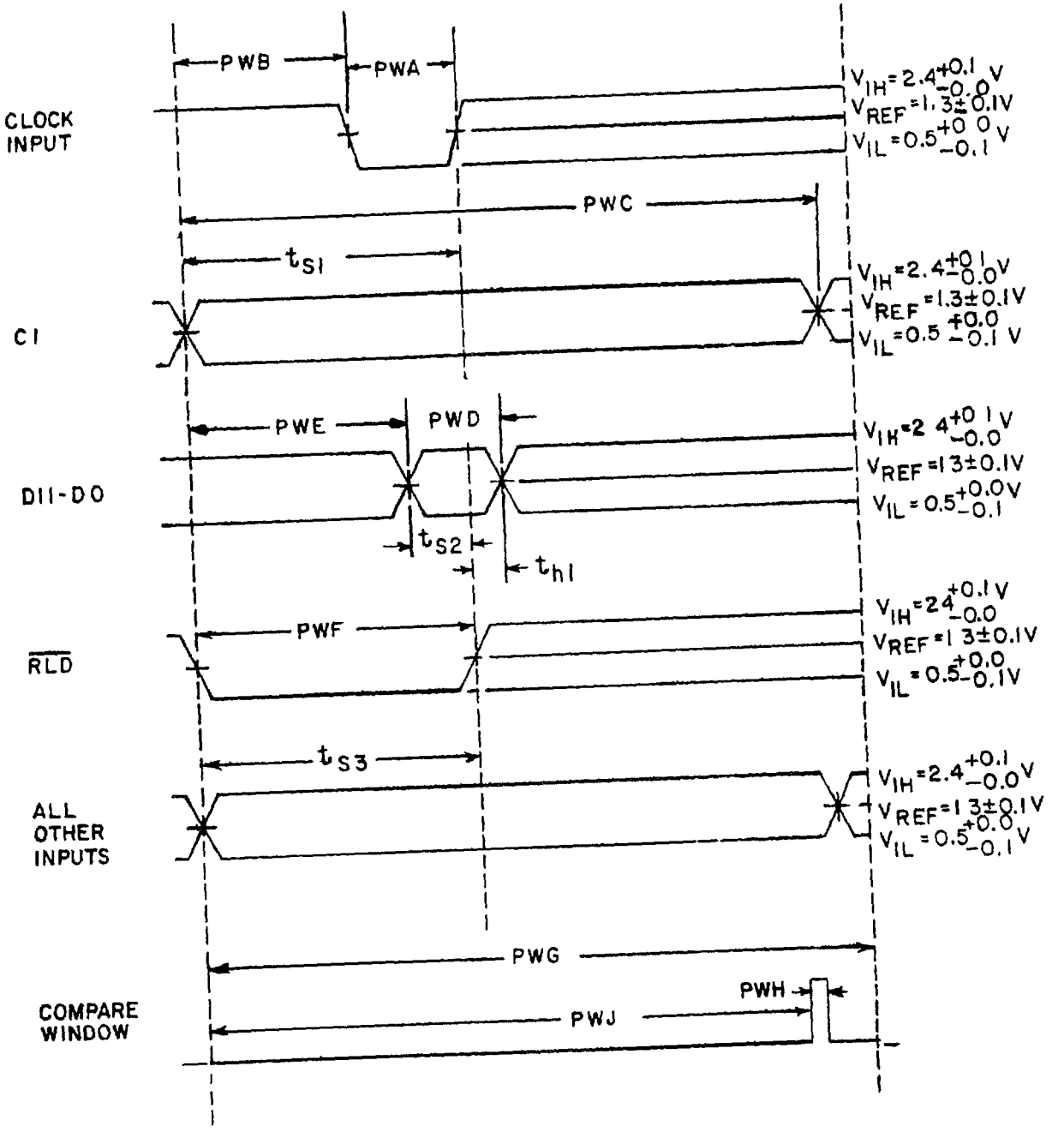
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:34Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters
PWA = 60 ±1 ns.
PWB = 75 ±1 ns.
PWC = 144 ±1 ns.
PWD = 10 ±1 ns.
PWE = 120 ±1 ns.

FIGURE 19. Dynamic functional test waveforms (N7, N8 and FPAT2) for table XV.

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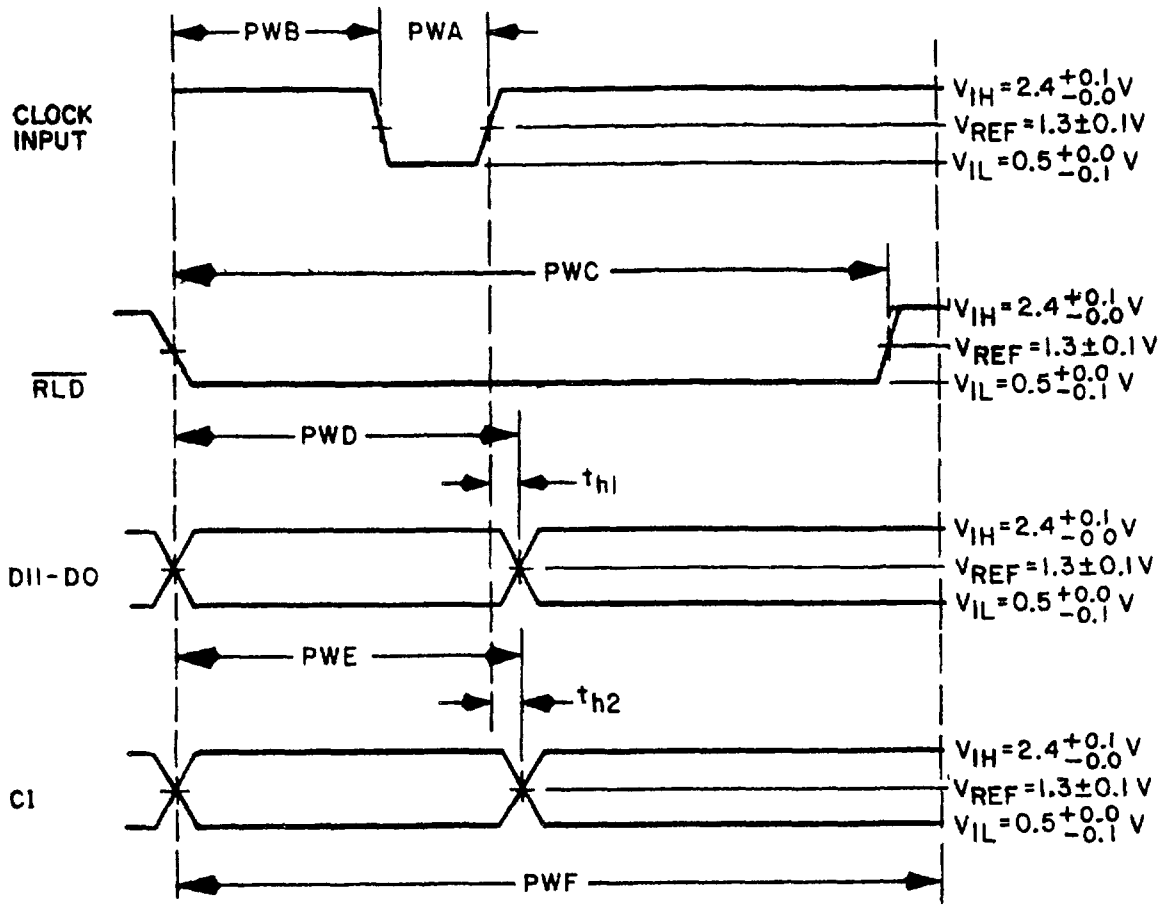


NOTE: Input pulse parameters.

PWA	= 70 ±1 ns.	PWH	= 13 ±1 ns.
PWB	= 110 ±1 ns.	PWJ	= 380 ±1 ns.
PWC	= 400±1 ns.	t_{S1}	= 180 ±2 ns.
PWD	= 34 ±1 ns	t_{S2}	= 28 ±2 ns
PWE	= 152 ±1 ns	t_{S3}	= 180 ±2 ns
PWF	= 186 ±1 ns	t_{h1}	= 6 ±2 ns.
PWG	= 424 ±1 ns		

FIGURE 20. Dynamic functional test waveforms (N9, N10, FPAT4, FPAT5 and vectors 1-8) for table XV.

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NOTE: Input pulse parameters:
PWA = 70 ±1 ns.
PWB = 110 ±1 ns.
PWC = 400 ±1 ns.
PWD = 184 ±1 ns.
PWE = 185 ±1 ns.
PWF = 424 ±1 ns.
t_{h1} = 4 ±2 ns.
t_{h2} = 5 ±2 ns.

FIGURE 21. Dynamic functional test waveforms (N9, N10, FPAT4, FPAT5 and vectors 9-34) for table XV.

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TABLE XVI. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	Connect G to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
P1	Y0 Transition time (t_{TLH}), C_L = 50 pF	14,27	28,29, 32,33	31	Open	Open	Open	13	1	17	ns	22,23
P2	Y1	16,27						15				
P3	Y2	18,27						17				
P4	Y3	20,27						19				
P5	Y4	27,39						40				
P6	Y5	27,37						38				
P7	Y6	24,27						23				
P8	Y7	22,27						21				
P9	Y8	2,27						3				
P10	Y9	4,27						5				
P11	Y10	6,27						7				
P12	Y11	8,27						9				
P13	$\overline{PL}$	27						35				
P14	$\overline{MAP}$	27						34				
P15	$\overline{VECT}$	28,32	27,29, 33					36				
P16	$\overline{FULL}$	26	12,27, 28,32	11	31,33	29		25				22,24

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TABLE XVII. Transition time (t_{THL}) test.

Test condition	Parameter	Connect: A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Initial limits 25°±5°C ambient: Min	Max	Unit	Test figure
Q1	Y0 Transition time (t_{THL}), C_L = 50 pF	14,27	28,29, 32,33	31	Open	Open	Open	13	1	10	ns	22,23
Q2	Y1	16,27						15				
Q3	Y2	18,27						17				
Q4	Y3	20,27						19				
Q5	Y4	27,39						40				
Q6	Y5	27,37						38				
Q7	Y6	24,27						23				
Q8	Y7	22,27						21				
Q9	Y8	2,27						3				
Q10	Y9	4,27						5				
Q11	Y10	6,27						7				
Q12	Y11	8,27						9				
Q13	$\overline{PL}$	27						35				
Q14	$\overline{MAP}$	27						34				
Q15	$\overline{VECT}$	28	27,29, 33	31,32				36				
Q16	$\overline{FULL}$	26	12,27, 28,32	11	31,33	29		25				22,24

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TABLE XVIII. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect: A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Initial limits 25°±5°C ambient: Min	Max	Unit	Test figure
R1	D0 to Y0 Propagation delay time (t_{PLH}), C_L = 50 pF	27,31	28,29, 32,33	14	Open	Open	14	13	2	18	ns	22,23
R2	D1 to Y1			16			16	15				
R3	D2 to Y2			18			18	17				
R4	D3 to Y3			20			20	19				
R5	D4 to Y4			39			39	40				
R6	D5 to Y5			37			37	38				
R7	D6 to Y6			24			24	23				
R8	D7 to Y7			22			22	21				
R9	D8 to Y8			2			2	3				
R10	D9 to Y9			4			4	5				
R11	D10 to Y10			6			6	7				
R12	D11 to Y11			8			8	9				
R13	I1 to Y0	14,27		31			31	13	5	52		
R14	I1 to Y1	16,27						15				
R15	I1 to Y2	18,27						17				
R16	I1 to Y3	20,27						19				
R17	I1 to Y4	27,39						40				
R18	I1 to Y5	27,37						38				
R19	I1 to Y6	24,27						23				
R20	I1 to Y7	22,27						21				

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TABLE XVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect: A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Initial limits 25°±5°C ambient: Min	Max	Unit	Test figure
R21	I1 to Y8 Propagation delay time (t_{PLH}), C_L = 50 pF	2,27	28,29, 32,33	31	Open	Open	31	3	5	52	ns	22,23
R22	I1 to Y9	4,27						5				
R23	I1 to Y10	6,27						7				
R24	I1 to Y11	8,27						9				
R25	I1 to $\overline{PL}$	27						35	3	37		
R26	I1 to $\overline{MAP}$	27						34	3	37		
R27	I1 to $\overline{VECT}$	28,32	27,29, 33					36	3	37		
	Setup (See fig. 22, note 4)	27,32, 33	26,28, 29,31	11			---	---	---	---	---	---
R28	$\overline{CC}$ to Y0	29,32	14,28, 31,33	27			27	13	3	30	ns	22,23
R29	$\overline{CC}$ to Y1		16,28, 31,33					15				
R30	$\overline{CC}$ to Y2		18,28, 31,33					17				
R31	$\overline{CC}$ to Y3		20,28, 31,33					19				
R32	$\overline{CC}$ to Y4		28,31, 33,39					40				
R33	$\overline{CC}$ to Y5		28,31 33,37					38				
R34	$\overline{CC}$ to Y6		24,28, 31,33					23				
R35	$\overline{CC}$ to Y7		22,28, 31,33					21				

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TABLE XVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect: A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Initial limits 25°±5°C ambient: Min	Max	Unit	Test figure
R36	$\overline{CC}$ to Y8 Propagation delay time (t_{PLH}), C_L = 50 pF	29,32	2,28, 31,33	27	Open	Open	27	3	3	30	ns	22,23
R37	$\overline{CC}$ to Y9		4,28, 31,33					5				
R38	$\overline{CC}$ to Y10		6,28, 31,33					7				
R39	$\overline{CC}$ to Y11		8,28, 31,33					9				
R40	$\overline{CCEN}$ to Y0	27,29, 32	14,31, 33	28			28	13		32		
R41	$\overline{CCEN}$ to Y1		16,31 33					15				
R42	$\overline{CCEN}$ to Y2		18,31, 33					17				
R43	$\overline{CCEN}$ to Y3		20,31, 33					19				
R44	$\overline{CCEN}$ to Y4		31,33, 39					40				
R45	$\overline{CCEN}$ to Y5		31,33, 37					38				
R46	$\overline{CCEN}$ to Y6		24,31, 33					23				
R47	$\overline{CCEN}$ to Y7		22,31, 33					21				
R48	$\overline{CCEN}$ to Y8		2,31, 33					3				
R49	$\overline{CCEN}$ to Y9		4,31, 33					5				

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TABLE XVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect: A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Initial limits 25°±5°C ambient: Min	Max	Unit	Test figure
R50	$\overline{CCEN}$ to Y10 Propagation delay time (t_{PLH}), C_L = 50 pF	27,29, 32	6,31, 33	28	Open	Open	28	7	3	32	ns	22,23
R51	$\overline{CCEN}$ to Y11	27,29, 32	8,31, 33	28	Open	Open	28	9	3	32	ns	22,23
R52	CP to $\overline{FULL}$	26	12,27, 28,32	11	31,33	29	11	25	4	42	ns	22,24
	Setup	12,26, 27	28,29, 31,32, 33,14, 16,18, 20,39, 37,24, 22,2, 4,6,8		Open	Open	Open	Open	---	---	---	---
R53	Clock → Y0	12,26, 27,29	28,31, 32,33, 14,16, 18,20, 39,37, 24,22, 2,4,6, 8				11	13	3	38	ns	22,25
R54	Clock → Y1							15				
R55	Clock → Y2							17				
R56	Clock → Y3							19				
R57	Clock → Y4							40				
R58	Clock → Y5							38				

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TABLE XVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect: A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Initial limits 25°±5°C ambient: Min	Max	Unit	Test figure
R59	Clock → Y6 Propagation delay time (t_{PLH}), C_L = 50 pF	12,26 27,29	28,31, 32,33, 14,16, 18,20, 39,37, 24,22, 2,4,6, 8	11	Open	Open	11	23	3	38	ns	22,25
R60	Clock → Y7							21				
R61	Clock → Y8							3				
R62	Clock → Y9							5				
R63	Clock → Y10							7				
R64	Clock → Y11							9				

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TABLE XIX. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect: A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Initial limits 25°±5°C ambient: Min	Max	Unit	Test figure
S1	D0 to Y0 Propagation delay time (t_{PHL}), C_L = 50 pF	27,31	28,29, 32,33	14	Open	Open	14	13	2	18	ns	22,23
S2	D1 to Y1			16			16	15				
S3	D2 to Y2			18			18	17				
S4	D3 to Y3			20			20	19				
S5	D4 to Y4			39			39	40				
S6	D5 to Y5			37			37	38				
S7	D6 to Y6			24			24	23				
S8	D7 to Y7			22			22	21				
S9	D8 to Y8			2			2	3				
S10	D9 to Y9			4			4	5				
S11	D10 to Y10			6			6	7				
S12	D11 to Y11			8			8	9				
S13	I1 to Y0	14,27		31			31	13	5	52		
S14	I1 to Y1	16,27						15				
S15	I1 to Y2	18,27						17				
S16	I1 to Y3	20,27						19				
S17	I1 to Y4	27,39						40				
S18	I1 to Y5	27,37						38				
S19	I1 to Y6	24,27						23				
S20	I1 to Y7	22,27						21				

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TABLE XIX. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect							Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
S21	I1 to Y8 Propagation delay time (t_{PHL}), C_L = 50 pF	2,27	28,29, 32,33	31	Open	Open	31	3	5	52	ns	22,23
S22	I1 to Y9	4,27						5				
S23	I1 to Y10	6,27						7				
S24	I1 to Y11	8,27						9				
S25	I1 to $\overline{PL}$	27						35	3	37		
S26	I1 to $\overline{MAP}$	27						34	3	37		
S27	I1 to $\overline{VECT}$	28,32	27,29, 33					36	3	37		
	Setup (See fig. 22, note 4)	27,32, 33	26,28, 29,31	11			---	---	---	---	---	---
S28	$\overline{CC}$ to Y0	29,32	14,28, 31,33	27			27	13	3	30	ns	22,23
S29	$\overline{CC}$ to Y1		16,28, 31,33					15				
S30	$\overline{CC}$ to Y2		18,28, 31,33					17				
S31	$\overline{CC}$ to Y3		20,28, 31,33					19				
S32	$\overline{CC}$ to Y4		28,31, 33,39					40				
S33	$\overline{CC}$ to Y5		28,31 33,37					38				
S34	$\overline{CC}$ to Y6		24,28, 31,33					23				
S35	$\overline{CC}$ to Y7		22,28, 31,33					21				

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TABLE XIX. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect: A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Initial limits 25°±5°C ambient: Min	Max	Unit	Test figure
S36	$\overline{CC}$ to Y8 Propagation delay time (t_{PHL}), C_L = 50 pF	29,32	2,28, 31,33	27	Open	Open	27	3	3	30	ns	22,23
S37	$\overline{CC}$ to Y9		4,28, 31,33					5				
S38	$\overline{CC}$ to Y10		6,28, 31,33					7				
S39	$\overline{CC}$ to Y11		8,28, 31,33					9				
S40	$\overline{CC}\overline{FN}$ to Y0	27,29, 32	14,31, 33	28			28	13		32		
S41	$\overline{CCEN}$ to Y1		16,31 33					15				
S42	$\overline{CCEN}$ to Y2		18,31, 33					17				
S43	$\overline{CCEN}$ to Y3		20,31, 33					19				
S44	$\overline{CCEN}$ to Y4		31,33, 39					40				
S45	$\overline{CCEN}$ to Y5		31,33, 37					38				
S46	$\overline{CCEN}$ to Y6		24,31, 33					23				
S47	$\overline{CCEN}$ to Y7		22,31, 33					21				
S48	$\overline{CCEN}$ to Y8		2,31, 33					3				
S49	$\overline{CCEN}$ to Y9		4,31, 33					5				

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TABLE XIX. Propagation delay time (t_{PHL}) test - Continued.

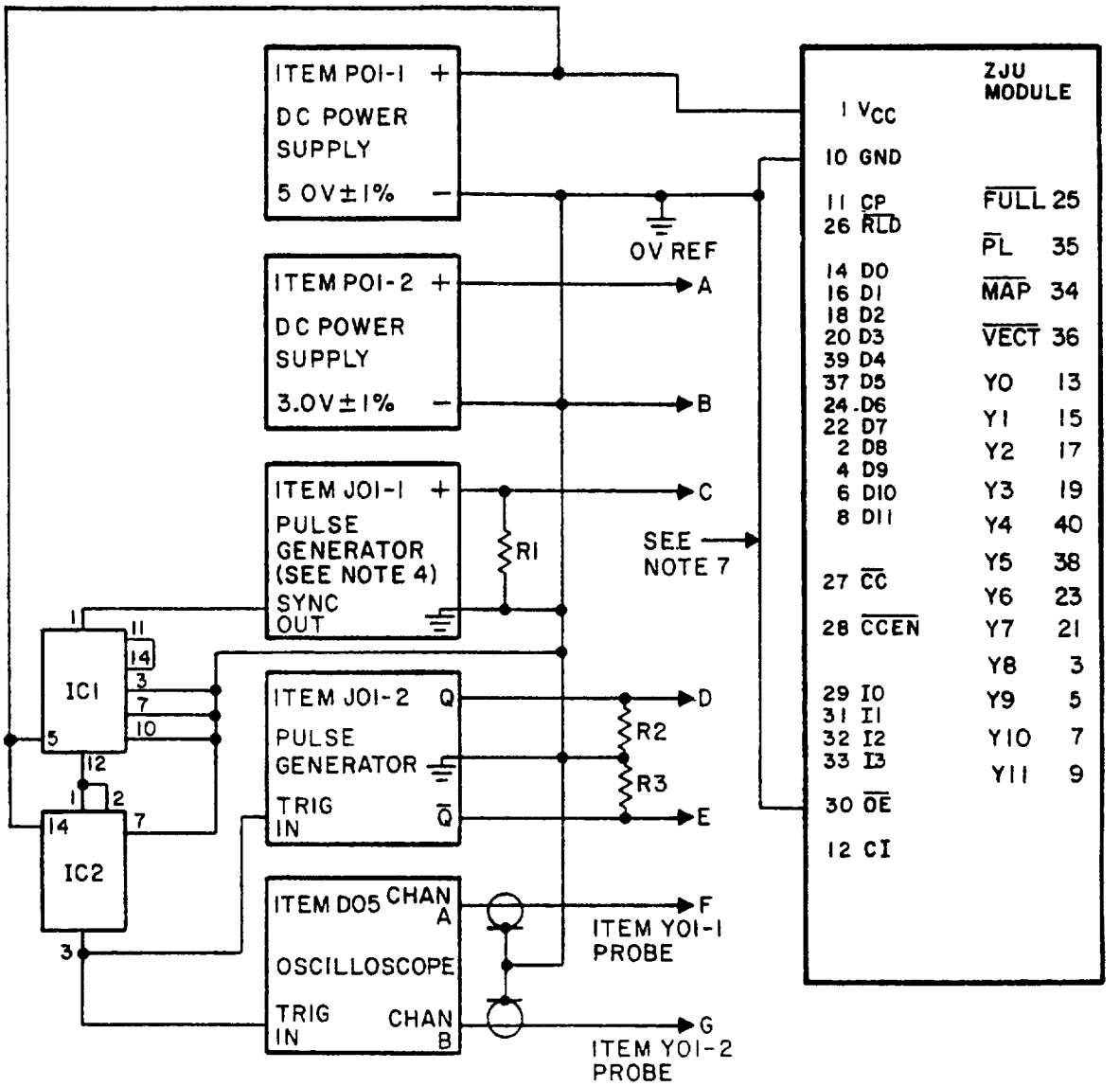
Test condition	Parameter	Connect: A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Initial limits 25°±5°C ambient: Min	Max	Unit	Test figure
S50	$\overline{CCEN}$ to Y10 Propagation delay time (t_{PHL}), C_L = 50 pF	27,29, 32	6,31, 33	28	Open	Open	28	7	3	32	ns	22,23
S51	$\overline{CCEN}$ to Y11	27,29, 32	8,31, 33	28	Open	Open	28	9	3	32	ns	22,23
S52	CP to $\overline{FULL}$	26	12,27, 28,32	11	31,33	29	11	25	4	42	ns	22,24
	Setup	12,26, 27	28,29, 31,32, 33,14, 16,18, 20,39, 37,24, 22,2, 4,6,8		Open	Open	Open	Open	---	---	---	---
S53	Clock → Y0	12,26, 27,29	28,31, 32,33, 14,16, 18,20, 39,37, 24,22, 2,4,6, 8				11	13	3	38	ns	22,25
S54	Clock → Y1							15				
S55	Clock → Y2							17				
S56	Clock → Y3							19				
S57	Clock → Y4							40				
S58	Clock → Y5							38				

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TABLE XIX. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect							Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
S59	Clock → Y6 Propagation delay time (t_{PHL}), C_L = 50 pF	12,26 27,29	28,31, 32,33, 14,16, 18,20, 39,37, 24,22, 2,4,6, 8	11	Open	Open	11	23	3	38	ns	22,25
S60	Clock → Y7							21				
S61	Clock → Y8							3				
S62	Clock → Y9							5				
S63	Clock → Y10							7				
S64	Clock → Y11							9				

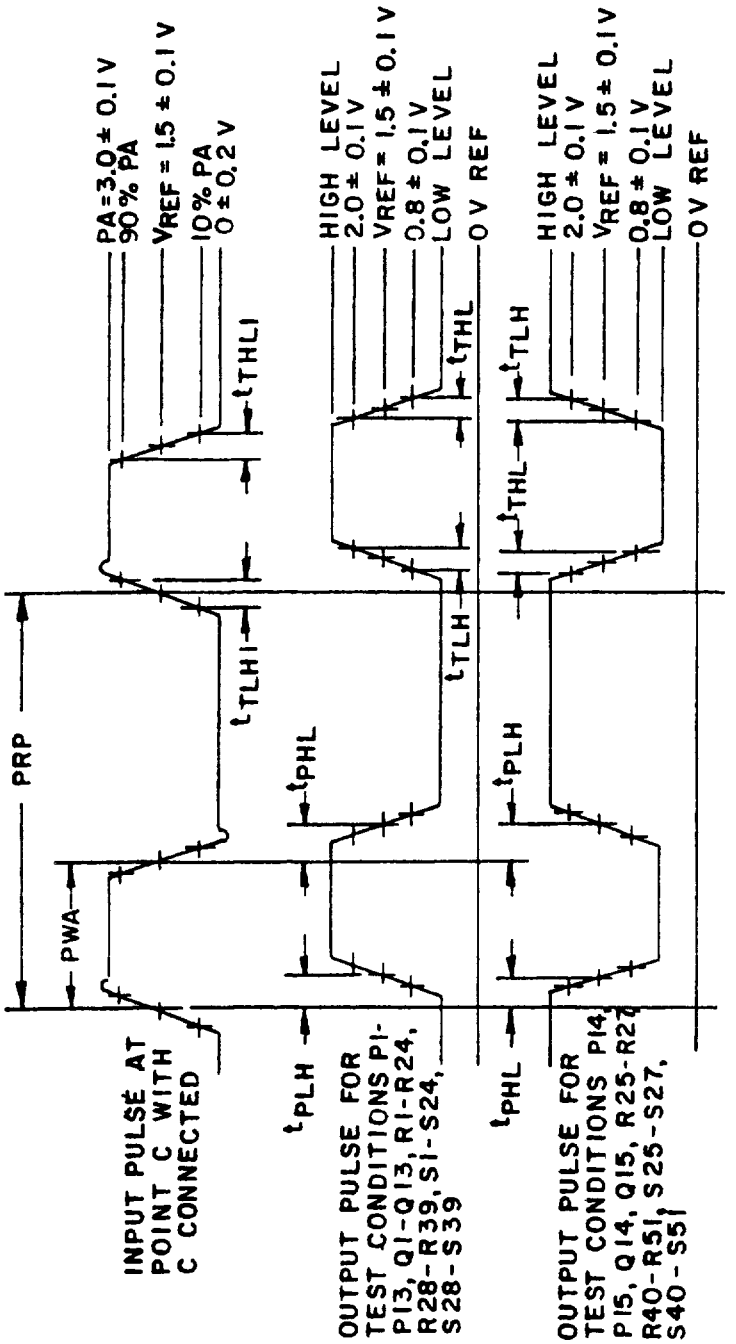
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NOTES:
1. R1-R3 = 51Ω ±5%, 1/2 W.
2. IC1 = 5490 or equivalent.
3. IC2 = 5400 or equivalent.
4. Item J01-1 setup· Switch to single pulse mode momentarily and apply single pulse to pin 11 after points A and B have been connected.
5 Load all outputs per figure 13.
6. Test conditions R28-R52 and S28-S52 shall be done in sequence after the setup conditions are made.
7. Disconnect pin 30 from ground for test conditions P13 and Q13 only. Pin 30 is grounded for all remaining test

FIGURE 22. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XVI, XVII, XVIII and XIX.

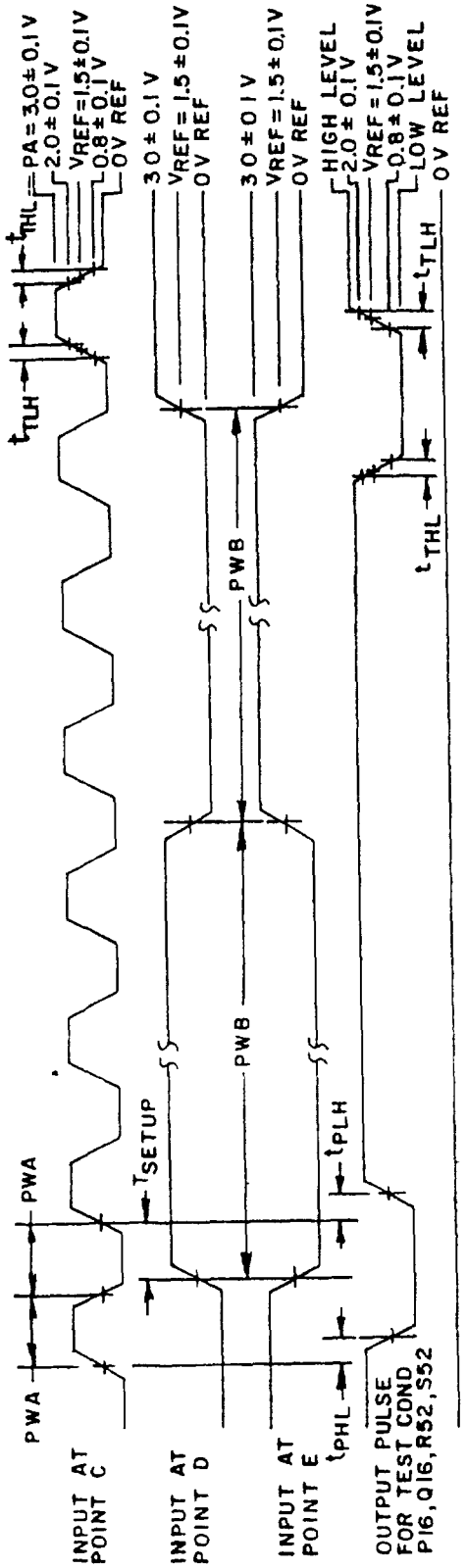
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NOTES:
1. Input pulse parameters
   PWA = 500 ±50 ns.
   PRP = 1000 ±100 ns.
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
2. Before making the specified measurements, verify that the low level output voltage is ≤ 0.5 V and the high level is ≥ 2.4 V.

FIGURE 23. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XVI, XVII, XVIII and XIX.

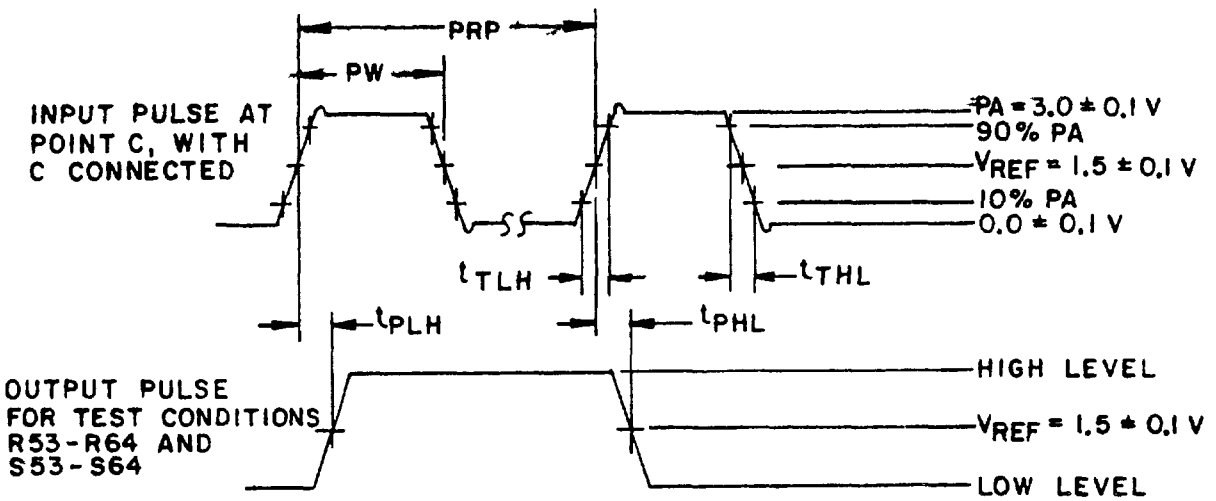
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NOTE: Input pulse parameters:
PWA = 500 ±50 ns.
PWB = 5000 ±250 ns.
T_{SETUP} = 400 ±40 ns.
$t_{TLH} = t_{THL}$ = 10 ±2 ns.

FIGURE 24 Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XVI, XVII, XVIII and XIX.

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NOTES.
1. Input pulse parameters:
   PW = 55 ±5 ns.
   PRP = 110 ±10 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Before making the specified measurements, verify that the low level is $\leq$ 0.5 V and the high level is $\geq$ 2.4 V.

FIGURE 25. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XVIII and XIX.

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TABLE XX. Disable delay time (t_{PLZ}) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
T1	$\overline{OE}$ to Y0 Disable delay time (t_{PLZ}), C_L = 50 pF	13	5	55	ns	26,27
T2	$\overline{OE}$ to Y1	15				
T3	$\overline{OE}$ to Y2	17				
T4	$\overline{OE}$ to Y3	19				
T5	$\overline{OE}$ to Y4	40				
T6	$\overline{OE}$ to Y5	38				
T7	$\overline{OE}$ to Y6	23				
T8	$\overline{OE}$ to Y7	21				
T9	$\overline{OE}$ to Y8	3				
T10	$\overline{OE}$ to Y9	5				
T11	$\overline{OE}$ to Y10	7				
T12	$\overline{OE}$ to Y11	9				

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TABLE XXI. Disable delay time (t_{PHZ}) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
U1	$\overline{OE}$ to Y0 Disable delay time (t_{PHZ}), C_L = 50 pF	13	6	60	ns	26,27
U2	$\overline{OE}$ to Y1	15	↓	↓	↓	↓
U3	$\overline{OE}$ to Y2	17	↓	↓	↓	↓
U4	$\overline{OE}$ to Y3	19	↓	↓	↓	↓
U5	$\overline{OE}$ to Y4	40	↓	↓	↓	↓
U6	$\overline{OE}$ to Y5	38	↓	↓	↓	↓
U7	$\overline{OE}$ to Y6	23	↓	↓	↓	↓
U8	$\overline{OE}$ to Y7	21	↓	↓	↓	↓
U9	$\overline{OE}$ to Y8	3	↓	↓	↓	↓
U10	$\overline{OE}$ to Y9	5	↓	↓	↓	↓
U11	$\overline{OE}$ to Y10	7	↓	↓	↓	↓
U12	$\overline{OE}$ to Y11	9	↓	↓	↓	↓

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TABLE XXII. Enable recovery time (t_{PZL}) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
V1	$\overline{OE}$ to Y0 Enable recovery time (t_{PZL}), C_L=50 pF	13	5	55	ns	26,27
V2	$\overline{OE}$ to Y1	15	↓	↓	↓	↓
V3	$\overline{OE}$ to Y2	17	↓	↓	↓	↓
V4	$\overline{OE}$ to Y3	19	↓	↓	↓	↓
V5	$\overline{OE}$ to Y4	40	↓	↓	↓	↓
V6	$\overline{OE}$ to Y5	38	↓	↓	↓	↓
V7	$\overline{OE}$ to Y6	23	↓	↓	↓	↓
V8	$\overline{OE}$ to Y7	21	↓	↓	↓	↓
V9	$\overline{OE}$ to Y8	3	↓	↓	↓	↓
V10	$\overline{OE}$ to Y9	5	↓	↓	↓	↓
V11	$\overline{OE}$ to Y10	7	↓	↓	↓	↓
V12	$\overline{OE}$ to Y11	9	↓	↓	↓	↓

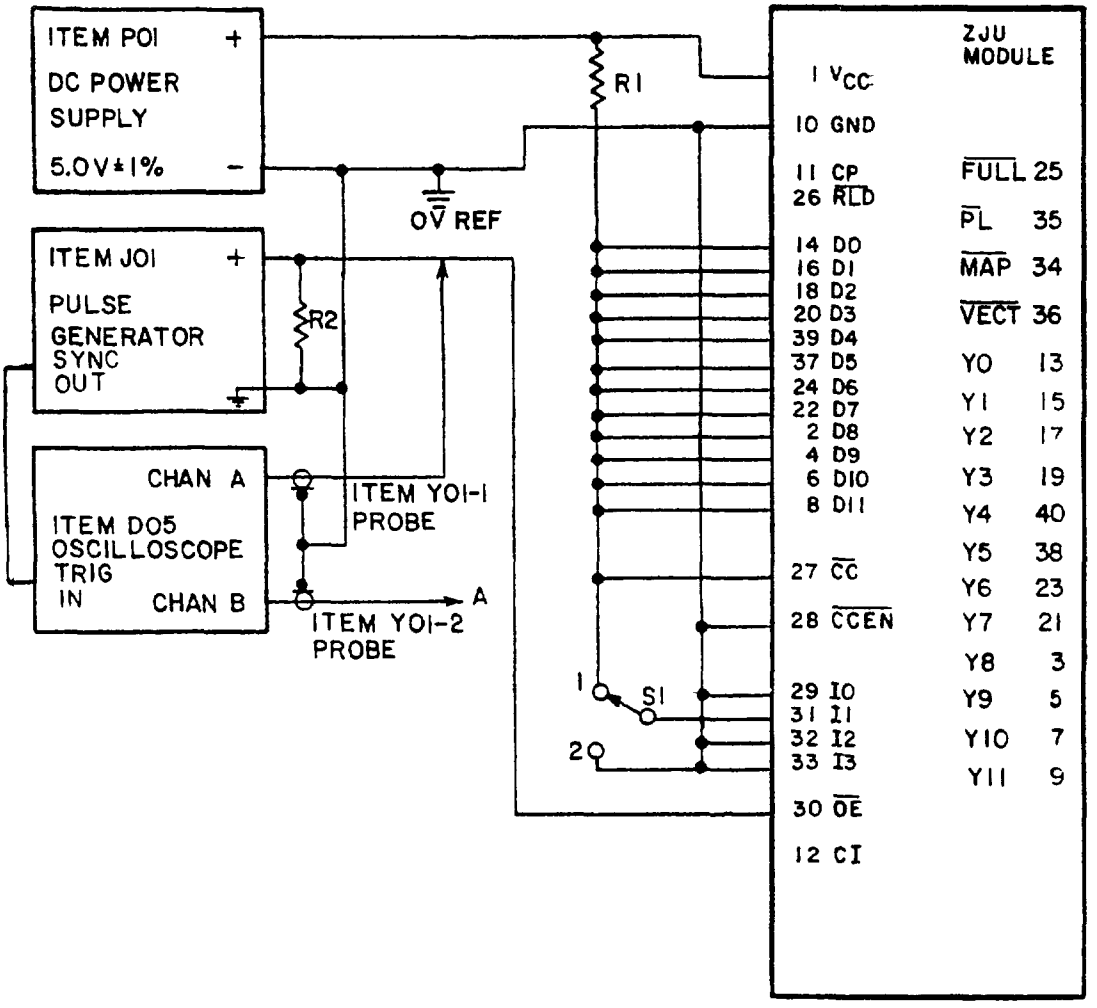
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TABLE XXIII. Enable recovery time (t_{PZH}) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
W1	$\overline{OE}$ to Y0	13	6	60	ns	26,27
W2	$\overline{OE}$ to Y1	15				
W3	$\overline{OE}$ to Y2	17				
W4	$\overline{OE}$ to Y3	19				
W5	$\overline{OE}$ to Y4	40				
W6	$\overline{OE}$ to Y5	38				
W7	$\overline{OE}$ to Y6	23				
W8	$\overline{OE}$ to Y7	21				
W9	$\overline{OE}$ to Y8	3				
W10	$\overline{OE}$ to Y9	5				
W11	$\overline{OE}$ to Y10	7				
W12	$\overline{OE}$ to Y11	9				

Enable recovery time (t_{PZH}), C_L=50 pF

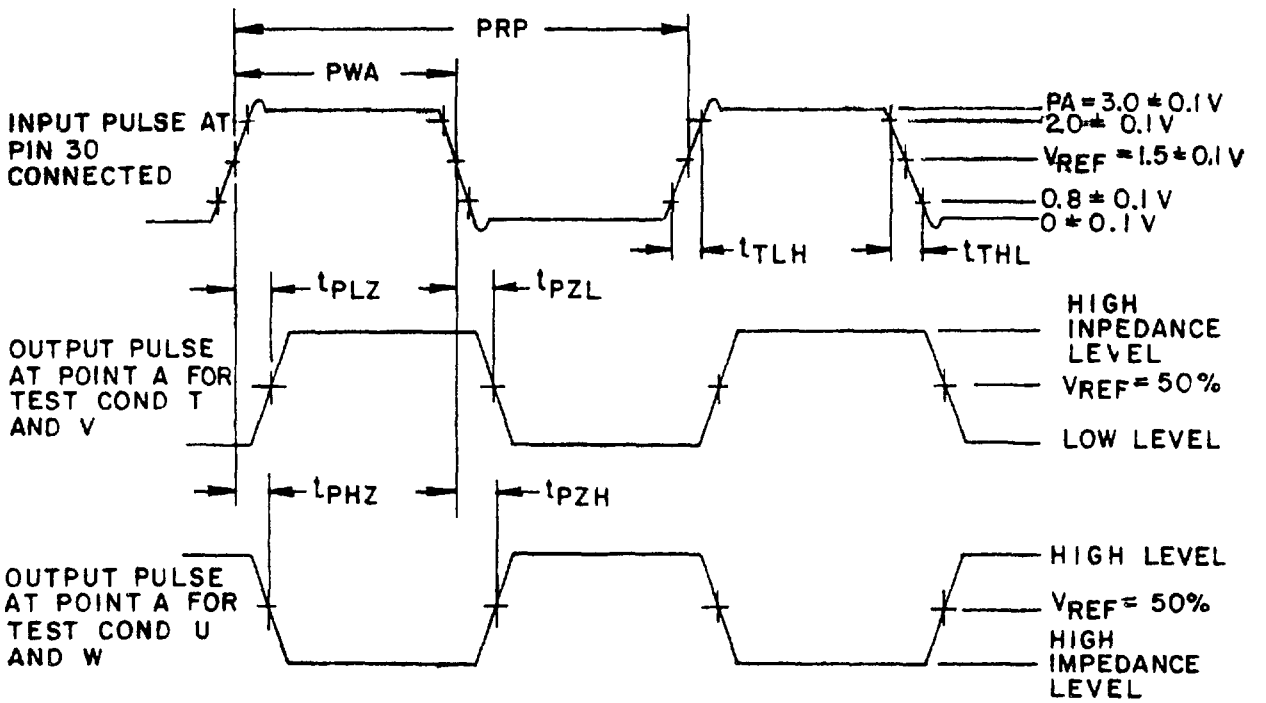
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NOTES
1. R1 = 1.0 kΩ ±5%, 1/4 W.
2. R2 = 51Ω ±5%, 1/2 W.
3. S1 switch = SPDT.
4. Load all outputs per figure 13.
5. For test conditions T and V, place switch S1 in position 2 and in position 1 for test conditions U and W.

FIGURE 26. Disable delay time (t_{PLZ} and t_{PHZ}) and enable recovery time (t_{PZL} and t_{PZH}) test circuit for tables XX, XXI, XXII and XXIII.

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NOTES.
1. Input pulse parameters:
   PWA = 500 ±50 ns
   PRP = 1000 ±100 ns.
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns.
2. Before making the specified measurements, verify that the output low level is $\leq$ 0.5 V and the high level is $\geq$ 2.4 V.

FIGURE 27. Disable delay time (t_{PLZ} and t_{PHZ}) and enable recovery time (t_{PZL} and t_{PZH}) test circuit waveforms for tables XX, XXI, XXII and XXIII.

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TABLE XXIV. Output crosstalk (low-level) test.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
X1	Y0	Output crosstalk, low level	27,16,18,20, 39,37,24,22, 2,4,6,8	14,29,32,33, 30,28	31	13	0	1,000	mV	28,29
X2	Y1		27,14,18,20, 39,37,24,22, 2,4,6,8	16,29,32,33, 30,28		15				
X3	Y2		27,14,16,20, 39,37,24,22, 2,4,6,8	18,29,32,33, 30,28		17				
X4	Y3		27,14,16,18, 39,37,24,22, 2,4,6,8	20,29,32,33, 30,28		19				
X5	Y4		27,14,16,18, 20,37,24,22, 2,4,6,8	39,29,32,33, 30,28		40				
X6	Y5		27,14,16,18, 20,39,24,22, 2,4,6,8	37,29,32,33, 30,28,		38				
X7	Y6		27,14,16,18, 20,39,37,22, 2,4,6,8	24,29,32,33, 30,28,		23				
X8	Y7		27,14,16,18, 20,39,37,24, 2,4,6,8	22,29,32,33, 30,28		21				
X9	Y8		27,14,16,18, 20,39,37,24, 22,4,6,8	2,29,32,33, 30,28		3				
X10	Y9		27,14,16,18, 20,39,37,24, 22,2,6,8	4,29,32,33, 30,28		5				

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TABLE XXIV. Output crosstalk (low-level) test - Continued

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
X11	Y10 Output crosstalk, low level	27,14,16,18, 20,39,37,24, 22,2,4,8	6,29,32,33, 30,28	31	7	0	1,000	mV	28,29
X12	Y11	27,14,16,18, 20,39,37,24, 22,2,4,6	8,29,32,33, 30,28	31	9				
X13	$\overline{PL}$	28,29	27,30,31,32, 33	14,16,18,20, 39,37,24,22, 2,4,6,8	35				
X14	$\overline{MAP}$	28,31	27,29,30,32, 33	14,16,18,20, 39,37,24,22, 2,4,6,8	34				
X15	$\overline{VECT}$	28,31,32	27,29,30,33	14,16,18,20, 39,37,24,22, 2,4,6,8	36				
---	Setup	26,29	12,27,28,30, 31,32,33	11	Open	---	---	---	
X16	$\overline{FULL}$	28,31,32	27,29,33,30	14,16,18,20, 39,37,24,22, 2,4,6,8	25	0	1,000	mV	

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TABLE XXV Output crosstalk (high-level) test.

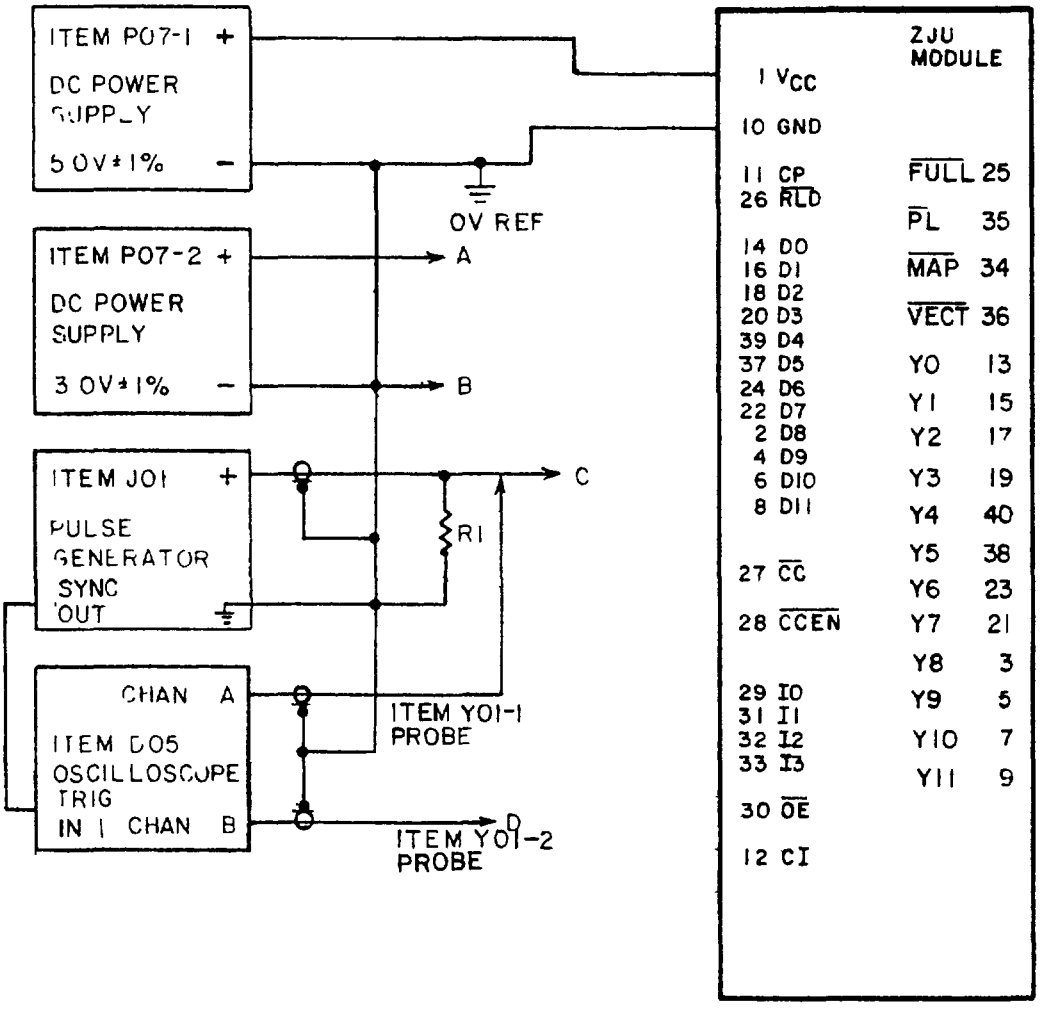
Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Y1	Y0	Output crosstalk, high level	28,14,31	27,29,33,30, 32	16,18,20,39, 37,24,22,2, 4,6,8	13	0	500	mV	28,29
Y2	Y1		28,16,31		14,18,20,39, 37,24,22,2,4, 6,8	15				
Y3	Y2		28,18,31		14,16,20,39, 37,24,22,2,4, 6,8	17				
Y4	Y3		28,20,31		14,16,18,39, 37,24,22,2,4, 6,8	19				
Y5	Y4		28,39,31		14,16,18,20, 37,24,22,2,4, 6,8	40				
Y6	Y5		28,37,31		14,16,18,20, 39,24,22,2,4, 6,8	38				
Y7	Y6		28,24,31		14,16,18,20, 39,37,22,2,4, 6,8	23				
Y8	Y7		28,22,31		14,16,18,20, 39,37,24,2,4, 6,8	21				
Y9	Y8		28,2,31		14,16,18,20, 39,37,24,22, 4,6,8	3				
Y10	Y9		28,4,31		14,16,18,20, 39,37,24,22, 2,6,8	5				

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TABLE XXV Output crosstalk (high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Y11	Y10 Output crosstalk, high level	28,6,31	27,29,33,30, 32	14,16,18,20, 39,37,24,22, 2,4,8	7	0	500	mV	28,29
Y12	Y11	28,8,31	27,29,33,30, 32	14,16,18,20, 39,37,24,22, 2,4,6	9				
Y13	$\overline{PL}$	28,31,32	27,29,33,30	14,16,18,20, 39,37,24,22, 2,4,6,8	35				
Y14	$\overline{MAP}$	28,29,31,32	27,33,30		34				
Y15	$\overline{VECT}$	28,29,31	27,32,33,30		36				
Y16	$\overline{FULL}$	28,31,32	27,29,33,30		25				

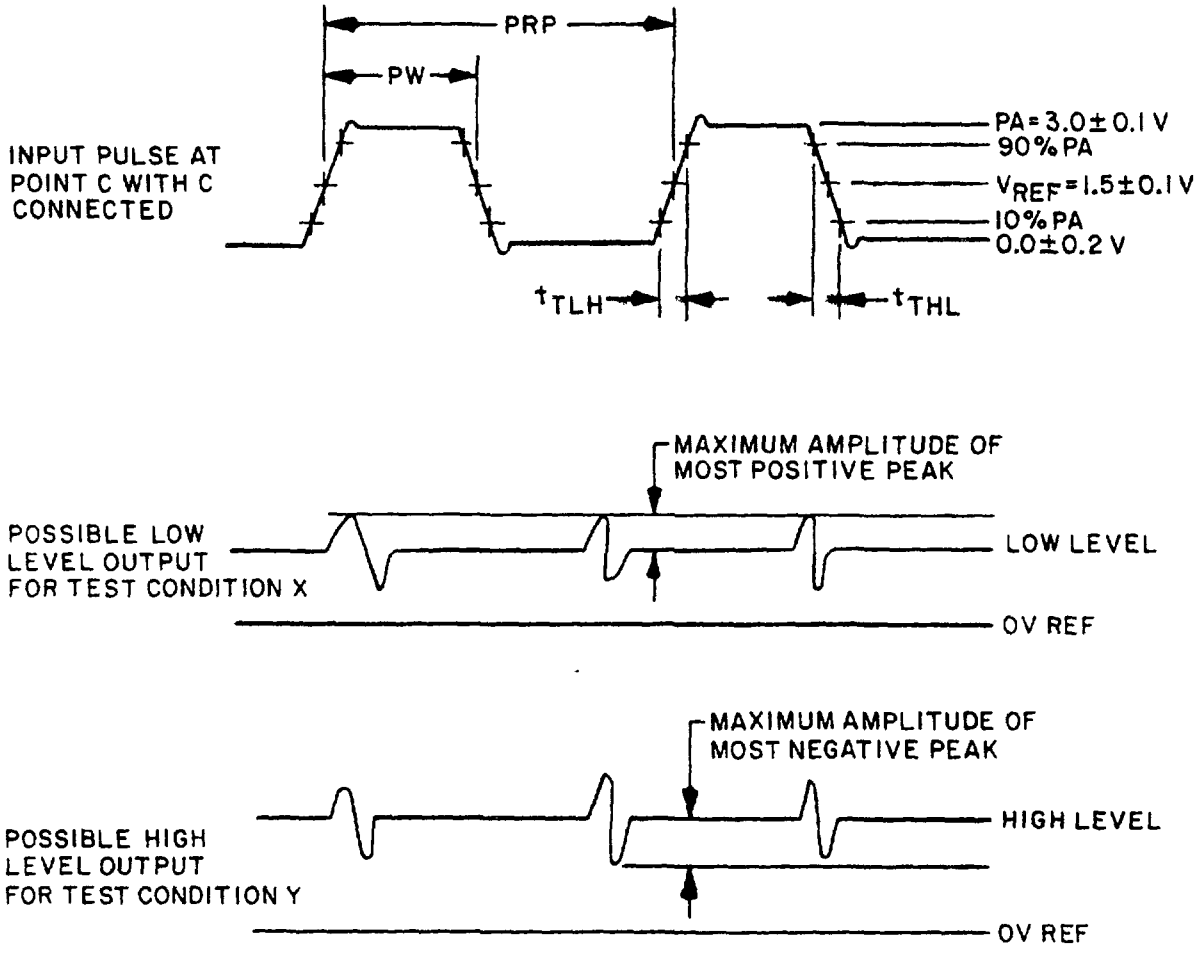
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NOTES
1. R1 = 51Ω ±5%, 1/2 W.
2. For setup condition prior to performing test condition X16, place item J01 in single pulse mode and pulse the module 5 times.
3. Load outputs per figure 13

FIGURE 28. Output crosstalk (low-level and high-level) test circuit for tables XXIV and XXV.

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NOTE: Input pulse parameters:
PW = 500 ±50 ns.
PRP = 1000 ±100 ns.
$t_{TLH} = t_{THL}$ = 15 ±3 ns.

FIGURE 29. Output crosstalk (low-level and high-level) test circuit waveforms for tables XXIV and XXV.

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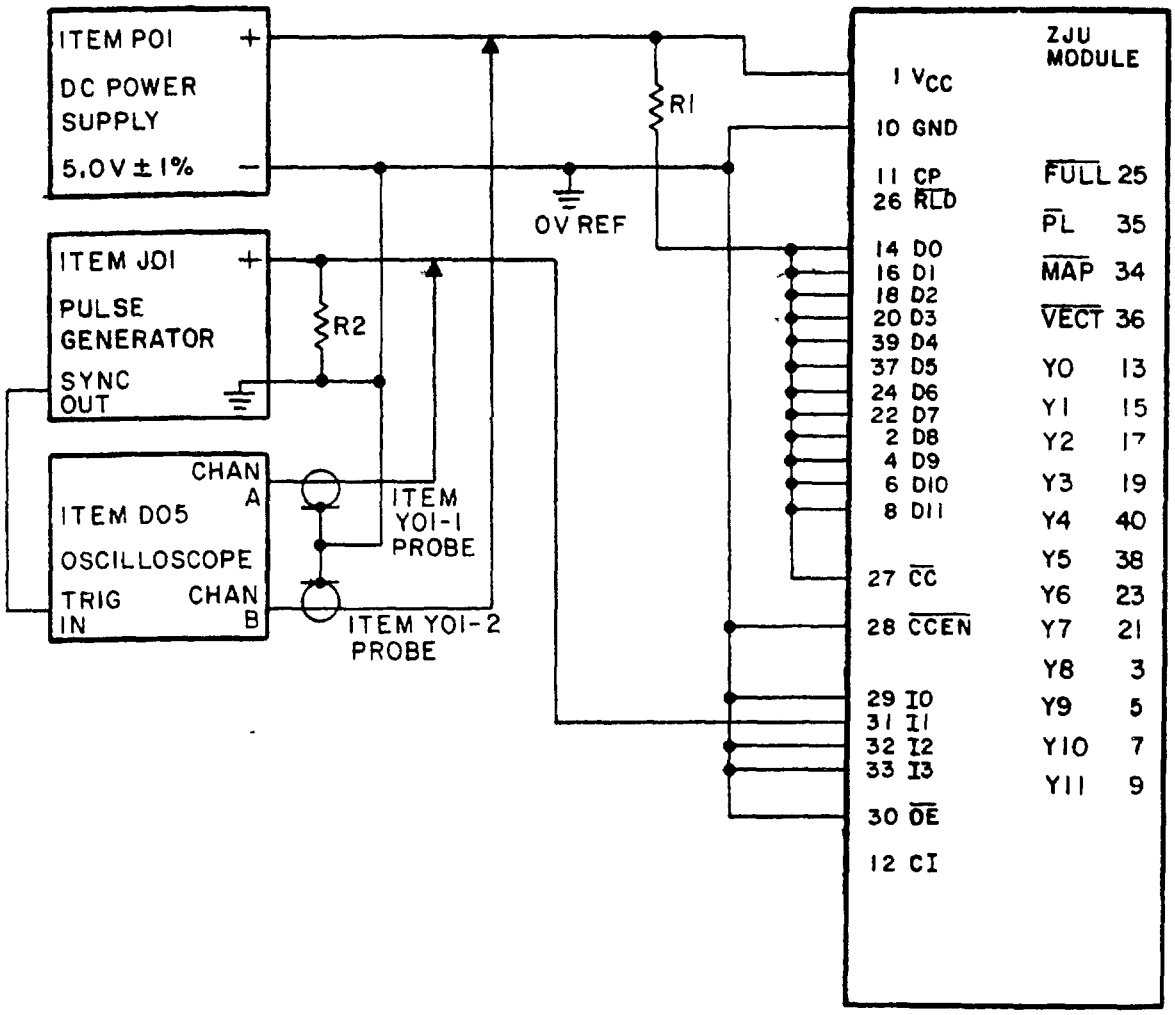
TABLE XXVI. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
Z	Power supply transient amplitude	0	400	mV	30,31

TABLE XXVII. Power supply filter capacitance test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
AA	Power supply filter capacitance	1	3.15	5.85	µF	32

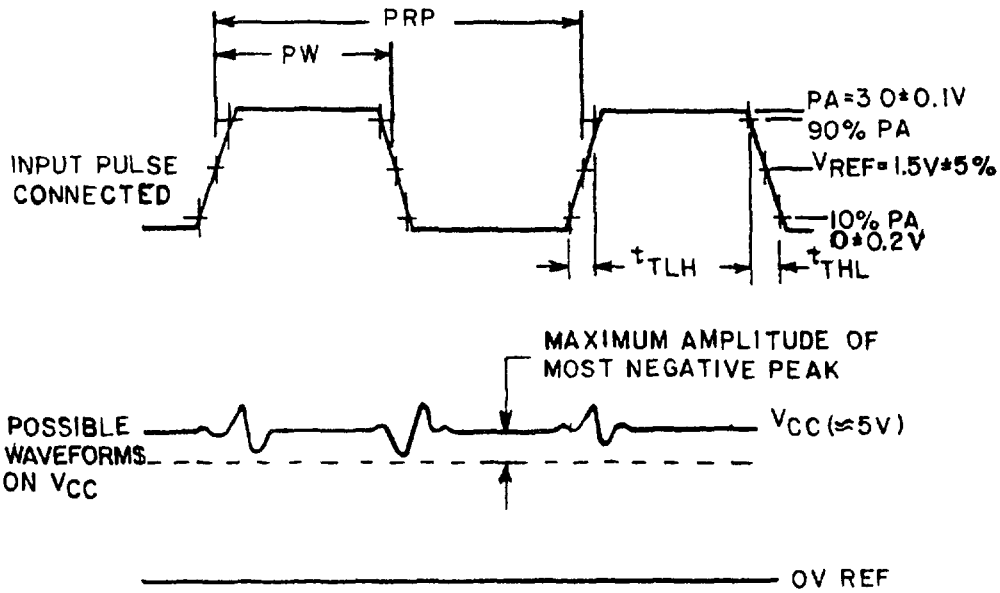
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NOTES.
1. R1 = 1.0 kΩ ±5%, 1/4 W.
2. R2 = 51Ω ±5%, 1/2 W.
3. Load all outputs per figure 13.

FIGURE 30. Power supply transient amplitude test circuit for table XXVI.

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NOTE: Input pulse parameters.
PW = 500 ±50 ns.
PRP = 1000 ±100 ns.
$t_{TLH} = t_{THL} = 10 \pm 2$ ns.

FIGURE 31. Power supply transient amplitude test circuit waveforms for table XXVI.

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TABLE XXVIII. Input capacitance test.

Test condition	Parameter	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
BB1	CP	11	2	30	pF	32
BB2	$\overline{RLD}$	26				
BB3	D0	14				
BB4	D1	16				
BB5	D2	18				
BB6	D3	20				
BB7	D4	39				
BB8	D5	37				
BB9	D6	24				
BB10	D7	22				
BB11	D8	2				
BB12	D9	4				
BB13	D10	6				
BB14	D11	8				
BB15	$\overline{CC}$	27				
BB16	$\overline{CCEN}$	28				
BB17	I0	29				
BB18	I1	31				
BB19	I2	32				
BB20	I3	33				
BB21	$\overline{OE}$	30				
BB22	CI	12				

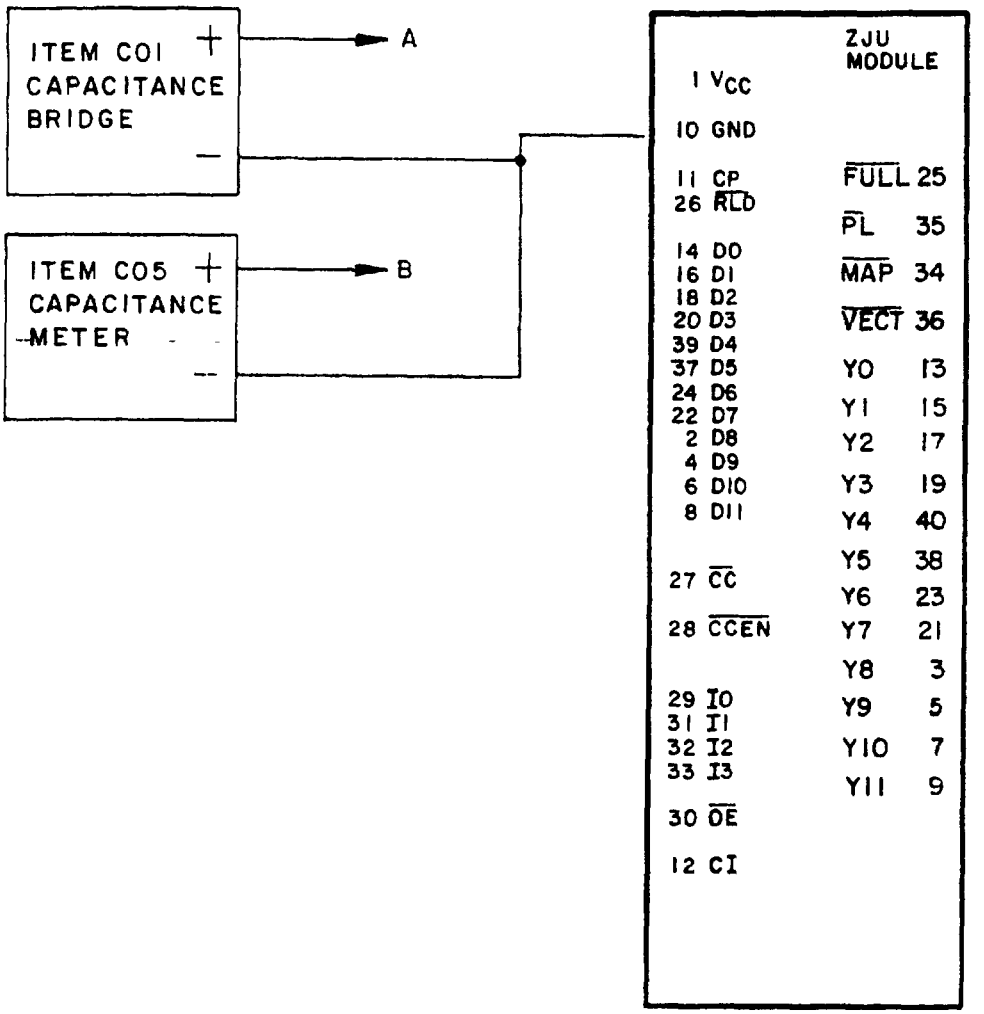
Input capacitance

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TABLE XXIX. Output capacitance test.

Test condition	Parameter	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
CC1	$\overline{\text{FULL}}$ Output capacitance	25	2	30	pF	32
CC2	$\overline{\text{PL}}$	35				
CC3	$\overline{\text{MAP}}$	34				
CC4	$\overline{\text{VECT}}$	36				
CC5	Y0	13				
CC6	Y1	15				
CC7	Y2	17				
CC8	Y3	19				
CC9	Y4	40				
CC10	Y5	38				
CC11	Y6	23				
CC12	Y7	21				
CC13	Y8	3				
CC14	Y9	5				
CC15	Y10	7				
CC16	Y11	9				

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NOTES.
1. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
2 Null out the effects of the test leads prior to connecting the module.
3. Capacitance bridge C01 setup:
   Frequency = 120 Hz
   V_{pk} = 0.3 V maximum.

FIGURE 32. Power supply filter capacitance, input and output capacitance test circuit for tables XXVII, XXVIII and XXIX.

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